

Standing-Carrier Upper Closure for the Neutrino Propagation Role Complex

Packet-image representation and mixed-source multiplicity beyond the two-anchor bound

Main result

PCR-5 fixes one terminal complete-carrier role and the SCIC quotient

$$R_5^{\text{term}} = \{r_{\text{car}}(T_{\nu}^{\text{osc}})\}, \quad \mathfrak{Q}_5^{\text{SCIC}} = A_5 / \sim_{\text{SCIC}}, \quad |\mathfrak{Q}_5^{\text{SCIC}}| \geq 2.$$

The analysis below gives a complete packet-image representation

$$\mathfrak{Q}_5^{\text{SCIC}} \cong \mathfrak{M}_5^{\text{stand}} := \bigcup_{F \in \mathfrak{F}_5^{\text{surv}}} \text{im}(\Xi_5 \circ \iota_F)$$

in which cross-family identifications are exactly complete-packet equalities. It also exhibits two PCR-4 HYB survivors: a phase-redundant mixed carrier with law/new-local deletion pair (1, 1) and a rank-split synergistic mixed carrier with pair (0, 0). Together with the PCR-5 anchor pairs (0, 1) and (1, 0), these imply

$$|\mathfrak{Q}_5^{\text{SCIC}}| \geq 4.$$

Thus neither an exact two-class nor an exact three-class standing-carrier quotient is compatible with the fixed target. The complete stratification of the general LAW, LOCAL, HID, COMP, SOL, and HYB packet images remains a separate classification problem.

Abstract

PCR-5 establishes universal compatible occupancy of the neutrino propagation role complex and a target-relative Standing Carrier Identity Closure (SCIC) quotient with at least two classes. The exact standing-carrier structure of that quotient is not fixed by the two benchmark classes alone. This paper develops a standing-faithful packet representation of the quotient and shows that mixed LAW-plus-singlet carriers already force a strictly larger lower bound.

The saturated SCIC evaluator algebra is reduced to a seven-type primitive generator system. Structural induction over the PCR-5 saturation grammar factors every admitted standing evaluator through the corresponding raw normal form, while the PCR-5 continuation generators induce a total fail-closed action on those coordinates. After standing-faithful descent, the seven-component packet

$$\Xi_5 = (T^{\text{face}}, R^{\text{stand}}, \Delta^{\text{del}}, C^{\text{act}}, M^{\text{match}}, B^{\text{bdy}}, F^{\text{same}})$$

satisfies

$$\Xi_5(x) = \Xi_5(y) \iff \Sigma_5^{\text{SCIC}}(x) = \Sigma_5^{\text{SCIC}}(y).$$

The PCR-4 family cover and canonical presentation maps therefore yield an exact packet-image representation of $\mathfrak{Q}_5^{\text{SCIC}}$.

The mixed-source calculation is pointwise. Two explicit HYB parameter objects are evaluated against the simultaneous PCR-4 conditions: target records, event-complete response, support and continuation, gauge and anomaly data, a common vacuum/deformation branch, oriented matching, activated-extension status, cross residuals, and a common cone. The phase-redundant point has deletion pair (1, 1), whereas the rank-split synergistic point has (0, 0). The two PCR-5 anchors have (0, 1) and (1, 0). Since the deletion coordinates are admitted standing-bearing no-repair consequences, the four points are pairwise SCIC-distinct and $|\mathfrak{Q}_5^{\text{SCIC}}| \geq 4$.

The result is exact at the level of quotient representation and identity, but it does not purport to furnish a complete stratum atlas for every symbolic family image. A full LAW/LOCAL/HID/COMP/SOL/HYB stratification, including all overlap maps, remains an open classification problem downstream of the theorems proved here.

Keywords. AASC; neutrino oscillation; SCIC; standing-carrier packet; packet-image representation; Weinberg operator; right-handed neutrino; mixed-source hybrid; four-class lower bound; carrier classification.

Scope and interpretation

The paper counts standing-carrier identities of one terminal role. It neither multiplies the role set nor asserts a particle–antiparticle relation. The principal conclusions are a four-class lower bound and an exact packet-image representation of the SCIC quotient; neither an exact finite cardinality nor a realized-carrier selection theorem is claimed.

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1 Problem, results, and proof structure

1.1 The question left open by PCR-5

The first five stages of the physical-carrier realization sequence separate role necessity, target interface, candidate generation, simultaneous survival, and compatible occupancy. PCR-5 establishes that every PCR-4 survivor has a complete compatible occupation and then forms the unique SCIC standing quotient [2–7]. Its strongest carrier statement is

$$|\mathfrak{Q}_5^{\text{SCIC}}| \geq 2.$$

That lower bound does not determine the structure of the quotient. In particular, it leaves open whether the two displayed anchors exhaust the standing classes, whether additional finite classes occur, or whether the quotient contains parameterized strata.

PCR-5A addresses two related but distinct tasks. The first is to represent the fixed SCIC quotient by a calculable packet map whose fibres are exactly the standing classes. The second is to classify the images of the PCR-4 survivor families into explicit strata. The results below complete the first task and sharpen the cardinal lower bound, while the second remains an explicit classification problem.

1.2 Three notions that must not be conflated

Definition 1.1 (Exact packet-image representation). *An exact packet-image representation is a classified map object M_5^{img} together with a canonical bijection*

$$M_5^{\text{img}} \cong \mathfrak{Q}_5^{\text{SCIC}},$$

where M_5^{img} is the union of the complete packet images of the PCR-4 survivor families fixed by the inherited theorem chain and equal packets are identified across family charts.

Definition 1.2 (Executed exact standing-carrier census). *An executed exact census is stronger. It consists of an exact packet-image representation together with a proved stratum atlas*

$$\{(\Theta_\alpha^{\text{surv}}, \Gamma_\alpha, \Xi_\alpha, \text{im } \Xi_\alpha, \text{Prov}_\alpha, \partial \text{Fail}_\alpha)\}_{\alpha \in I}$$

covering every family image, with the index set, parameter quotients, packet values or packet moduli, failure boundaries, and all cross-family overlap maps explicitly classified.

The distinction is mathematical rather than terminological: an exact representation identifies every quotient element and every equality correctly, whereas an executed census additionally resolves the internal stratum structure of the image.

1.3 Main theorems

Theorem 1.3 (Exact packet-image representation). *Let*

$$\mathfrak{M}_5^{\text{stand}} = \bigcup_{F \in \mathfrak{F}_5^{\text{surv}}} \text{im}(\Xi_5 \circ \iota_F), \quad \mathfrak{F}_5^{\text{surv}} = \{\text{LAW}, \text{LOCAL}, \text{HID}, \text{COMP}, \text{SOL}, \text{HYB}\}.$$

Then packet equality performs precisely SCIC equality and

$$\boxed{\mathfrak{M}_5^{\text{stand}} \cong \mathfrak{Q}_5^{\text{SCIC}}.}$$

This theorem is an exact representation theorem; it is not by itself an executed stratum classification of the six images.

Theorem 1.4 (Mixed-source four-class theorem). *There exist four PCR-4 survivors*

$$x_W, \quad x_{3N}, \quad x_{\text{red}}, \quad x_{\text{syn}}$$

with law/new-local deletion pairs

$$\mathbf{d}_{W \oplus 3N}(x_W) = (0, 1), \quad \mathbf{d}_{W \oplus 3N}(x_{3N}) = (1, 0), \quad \mathbf{d}_{W \oplus 3N}(x_{\text{red}}) = (1, 1), \quad \mathbf{d}_{W \oplus 3N}(x_{\text{syn}}) = (0, 0).$$

Consequently

$$|\mathfrak{Q}_5^{\text{SCIC}}| \geq 4,$$

and exact two and exact three are false.

Corollary 1.5 (Scope of the PCR-5A result). *The results establish:*

1. an exact packet-image representation of the fixed SCIC quotient;
2. an unconditional four-class lower bound inside that quotient;
3. an open classification problem for the complete family-stratification census.

A full PCR-6 characterization of the entire quotient therefore depends on completing the stratum atlas.

1.4 Dependency structure

The proof follows the fixed dependency order

PCR-3/PCR-4 survivor cover $\implies$ primitive-generator factorization $\implies$ standing-faithful packet,
 survivor-HYB typing $\implies$ explicit mixed PCR-4 witnesses $\implies |\mathfrak{Q}_5^{\text{SCIC}}| \geq 4$,
 family packet images $\implies$ exact packet-image representation,

while the further reduction

family packet images $\implies$ executed exact stratum atlas

remains open.

No selector and no census-by-definition

Simplicity, field count, naturalness, familiarity, and ultraviolet preference do not enter the equivalence relation. Likewise, writing a family image as $\text{im}(\Xi_5 \circ \iota_F)$ does not by itself classify that image into exact strata.

1.5 Position within the PCR sequence

Physical carrier characterization is meaningful only after the domain to be characterized has been fixed at the required level of resolution. PCR-5A therefore sits between compatible occupancy and carrier characterization: it represents the SCIC quotient, identifies mixed-source classes that enlarge the lower bound, and isolates the remaining family-stratification problem that must be solved before the entire quotient can be characterized stratum by stratum.

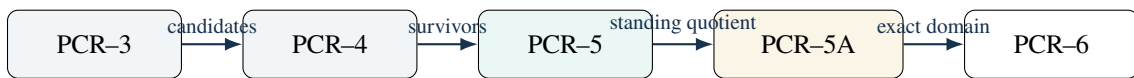


Figure 1: Position of PCR-5A in the physical-carrier realization sequence.

Stage	Mathematical task	Output used downstream
PCR-3	Generate every same-target source-coherent carrier presentation.	Candidate groupoid and carrier equivalence.
PCR-4	Impose every simultaneous target, consistency, and common-witness condition.	Exact survivor quotient A_5 .
PCR-5	Prove complete compatible role occupancy and form terminal standing identity.	$\mathfrak{Q}_5^{\text{SCIC}}$, with $ \mathfrak{Q}_5^{\text{SCIC}} \geq 2$.
PCR-5A	Represent the quotient faithfully, calibrate mixed classes, and classify the family images.	Exact representation and $ \mathfrak{Q}_5^{\text{SCIC}} \geq 4$; complete stratum atlas remains open.
PCR-6	Characterize the physical nature of every exact class or stratum.	Global characterization depends on the completed PCR-5A stratum atlas.

1.6 Why the mixed calculation is structurally decisive

The two PCR-5 anchors separate the presence and absence of one active new-local source kind. A mixed construction tests source composition itself. Three dispositions are possible: one source is standing-inert; both are independently sufficient; or both are jointly necessary. The last two are realized below by explicit PCR-4 points and cannot be collapsed to either single-source anchor because their no-repair deletion responses differ.

1.7 Strength of the result

The four-class theorem strengthens the PCR-5 lower bound independently of the unresolved general family stratification. The packet-image theorem is stronger than a model list because every PCR-4 survivor is represented and every SCIC equality is captured. It is nevertheless weaker than an executed atlas of the packet images. Keeping these two levels separate fixes the precise strength of the result.

2 Fixed target, candidate domain, and inherited theorem chain

2.1 Fixed target and one terminal role

The target remains the PCR-1/PCR-2 neutrino-oscillation target. It contains the charged-current source/detection boundary, the two nuisance-separated visible response-mode classes, the lawful support/continuation structure, and the target-admitted consistency, scale, and matching uses. PCR-5 separately declares one terminal complete-carrier role

$$R_5^{\text{term}} = \{r_{\text{car}}(T_{\nu}^{\text{osc}})\}.$$

PCR-5A changes neither object.

2.2 PCR-3 candidate completeness

PCR-3 proves that every same-target carrier architecture in the declared comparison class has a finite source-coherent hypergraph presentation, including arbitrary lawful finite hybrids and symbolic parameter domains [5]. Let $\tilde{\Theta}_{\text{F}}$ denote the corresponding family-presentation loci and $\approx_{\text{inst}}$ complete active carrier-instance equivalence.

2.3 PCR-4 exact survivor residue

PCR-4 evaluates those presentations with one simultaneous witness system. Its exact survivor quotient is

$$A_5 = \mathfrak{S}_v^{(4)} \cong (\Theta_{\text{LAW}}^{\text{surv}} \sqcup \Theta_{\text{LOCAL}}^{\text{surv}} \sqcup \Theta_{\text{COMP}}^{\text{surv}} \sqcup \Theta_{\text{SOL}}^{\text{surv}} \sqcup \Theta_{\text{HID}}^{\text{surv}} \sqcup \Theta_{\text{HYB}}^{\text{surv}}) / \sim_{\text{inst}}. \quad (1)$$

The strict BASE stratum is empty, while TOP, EXT, BDY, and ENV can occur only inside surviving hybrids. Equation (1) is a theorem-level cover, not a model catalogue [4].

2.4 PCR-5 standing identity

PCR-5 activates the target-complete standing witness family $\mathcal{W}_5^{\text{stand}}$, the total fail-closed continuation monoid $\mathcal{U}_5^{\text{stand}}$, and the signature

$$\Sigma_5^{\text{SCIC}}(x) = (w(u(x)))_{(w,u) \in \mathcal{W}_5^{\text{stand}} \times \mathcal{U}_5^{\text{stand}}}.$$

The standing relation is

$$x \sim_{\text{SCIC}} y \iff \Sigma_5^{\text{SCIC}}(x) = \Sigma_5^{\text{SCIC}}(y), \quad \mathfrak{Q}_5^{\text{SCIC}} = A_5 / \sim_{\text{SCIC}}.$$

Standing Carrier Identity Closure makes this the unique same-target terminal carrier quotient [6]. PCR-5 also proves

$$\Sigma_5^{\text{SCIC}}(x_W) \neq \Sigma_5^{\text{SCIC}}(x_{3N}).$$

Theorem 2.1 (No-backflow). *PCR-5A may calculate only the image and fibres of the fixed map $\Sigma_5^{\text{SCIC}} : A_5 \rightarrow \mathcal{Y}_5^{\text{stand}}$. It may not alter the target, role set, PCR-3 source ontology, PCR-4 survivor membership, PCR-5 occupancy theorem, or SCIC relation.*

Proof. Each listed object is fixed upstream of the census. Re-forming any of them from the desired cardinality would reverse the dependency order. A new probe, source kind, target record, or comparison regime is a successor target; an excluded PCR-4 point cannot be repopulated by PCR-5A. $\square$

2.5 Family presentation maps

For each principal family F , let

$$\iota_F : \Theta_F^{\text{surv}} \longrightarrow A_5$$

be the coproduct inclusion followed by the instance quotient in (1). The maps need not be injective; different source presentations can represent one inherited carrier.

Lemma 2.2 (Presentation-map coverage). *The images of the six maps ι_F cover A_5 . If two family presentations are instance-equivalent, their images in A_5 coincide.*

Proof. This is the evaluated-residue theorem and pointwise family form of PCR-4 specialized to the six principal surviving terms. The pure-stratum reductions are imposed before quotienting. $\square$

2.6 The target fingerprint in operational form

The fixed target does not begin from a mass matrix. Its operational anchors are the charged-current preparation/detection sectors, the event-complete propagation instrument, and at least two nuisance-separated visible phase-frequency classes. PCR-1 establishes the strict nesting of response multiplicity above nonalignment and excludes the strict massless-neutrino Standard Model from both roles. PCR-2 then fixes the complete target-facing trace. In the notation used here, that trace contains:

1. the charged-current source and detection record maps;
2. the event-complete active and null outcome instrument;

3. the visible response-mode classes together with their calibration and environment transports;
4. the source-to-detector continuation and causal/support data;
5. the gauge, anomaly, vacuum, deformation, scale, matching, and activated extension witnesses admitted by the target;
6. the one-completion joint-incidence data preventing witness splicing.

A completion can differ microscopically while retaining this target fingerprint. Conversely, a change in record semantics, admitted operator category, source/detector interaction, or background is a target update rather than a new point of the same census.

2.7 Exact PCR-3 source ontology

PCR-3 types each active carrier-bearing node by support, persistence, derivation status, constituent kind, intrinsic attributes, source parameter domain, and target attachment. Carrier relations include coupling, composition, emergence, projection/dilation, boundary attachment, fusion/junction, environmental co-carriage, and source-constraint sharing. The finite-hypergraph theorem retains higher-arity incidence and does not force every architecture into pairwise couplings. These details matter in PCR-5A because a third standing class can be carried by an incidence or continuation effect even when all constituent families are individually anchor-equivalent.

The candidate-completeness theorem is parametric. It covers noncompact, representation-valued, categorical, and otherwise symbolic source domains. Thus an exact PCR-5A result must be formulated over exact symbolic loci rather than a finite model register.

2.8 Exact PCR-4 survivor conditions

For each family F , PCR-4 constructs a global family map

$$\widehat{\Phi}_F : \Theta_F^{\text{part}} \longrightarrow \mathcal{Z}_4^{\text{all}}$$

and proves

$$\Theta_F^{\text{surv}} = \widehat{\Phi}_F^{-1}(\mathcal{Z}_4^{\text{sem}}).$$

The semantic locus simultaneously enforces target records, role capability, event completeness, active transfer, visible response multiplicity, locality/continuation, gauge attachment, anomaly closure, vacuum/deformation, matching, activated extensions, relation residuals, and common-cone existence. It is not the product of marginal pass sets.

For later reference, the principal loci are:

$$\begin{aligned} \Theta_{\text{LAW}}^{\text{surv}} &= \text{PCR-4 equation (51),} \\ \Theta_{\text{LOCAL}}^{\text{surv}} &= \text{PCR-4 equation (52),} \\ \Theta_{\text{COMP}}^{\text{surv}} &= \text{PCR-4 equation (53),} \\ \Theta_{\text{SOL}}^{\text{surv}} &= \text{PCR-4 equation (56),} \\ \Theta_{\text{HID}}^{\text{surv}} &= \text{PCR-4 equation (57),} \\ \Theta_{\text{HYB}}^{\text{surv}} &= \widehat{\Phi}_{\text{HYB}}^{-1}(\mathcal{Z}_4^{\text{sem}}). \end{aligned}$$

The last locus includes every cross-anomaly, cross-vacuum, mixed matching, common-cone, and global relation created by gluing.

Lemma 2.3 (One-point discipline). *Every PCR-5A packet is evaluated at one PCR-4 survivor point. A packet may not take its target channel from one parameter point, its anomaly cancellation from another, and its vacuum or matching map from a third.*

Proof. PCR-4 survivor membership is defined by one common witness and one source parameter point. The family maps ι_F land in the quotient of those complete points. Splicing values from several points does not define an element of A_5 . $\square$

2.9 SCIC activation and deletion standing

PCR-5 forms $\mathcal{W}_5^{\text{stand}}$ by lawful-redescription-invariant saturation of the PCR-2 target uses, PCR-4 survival relations, and PCR-5 role/compatibility/failure uses. In particular the kindwise no-repair deletion response is standing-bearing because its value is the preservation or destruction of the already fixed complete role packet, not the name of a deleted source component. This distinction is essential: a source label can be provenance-only, while the failure of a mandatory role path is a target-standing consequence.

Lemma 2.4 (Source labels do not enter by themselves). *Let $x, y \in A_5$ differ only in a raw source label or presentation coordinate. If every admitted evaluator after every licensed continuation agrees, then $x \sim_{\text{SCIC}} y$, irrespective of PCR-3 family ownership.*

Proof. This is the defining fibre condition of SCIC and the standing-faithful quotient used below. $\square$

3 Standing-packet representation of the SCIC quotient

3.1 Anchor fibres and residual domain

Set

$$s_W := \Sigma_5^{\text{SCIC}}(x_W), \quad s_{3N} := \Sigma_5^{\text{SCIC}}(x_{3N}).$$

PCR-5 proves $s_W \neq s_{3N}$. Define

$$\begin{aligned} C_W &= (\Sigma_5^{\text{SCIC}})^{-1}(s_W), & C_{3N} &= (\Sigma_5^{\text{SCIC}})^{-1}(s_{3N}), \\ A_5^{\text{third}} &= A_5 \setminus (C_W \cup C_{3N}), & \mathfrak{Q}_5^{\text{third}} &= A_5^{\text{third}} / \sim_{\text{SCIC}}. \end{aligned}$$

Theorem 3.1 (Formal anchor decomposition). *There is a canonical disjoint decomposition*

$$\mathfrak{Q}_5^{\text{SCIC}} \cong \{s_W, s_{3N}\} \sqcup \mathfrak{Q}_5^{\text{third}}.$$

Proof. The anchor fibres are two distinct fibres of the complete signature. Their complement is saturated under signature equality. Quotienting the partition gives the result. $\square$

3.2 Why source families are not quotient labels

LAW, LOCAL, HID, COMP, SOL, and HYB are source-generation and survivor-presentation classes. They can overlap, and one source family can contain several standing classes. Conversely, presentations from different families can determine the same complete standing packet. The standing quotient is therefore governed by packet equality rather than family nomenclature.

3.3 Exact packet-image representation

Once the packet Ξ_5 is proved complete, define

$$\mathcal{I}_F := \text{im}(\Xi_5 \circ \iota_F)$$

and

$$\mathfrak{M}_5^{\text{stand}} := \bigcup_{F \in \mathfrak{F}_5^{\text{surv}}} \mathcal{I}_F \subseteq \mathcal{Y}_5^{\Xi}. \quad (2)$$

The common packet codomain identifies cross-family overlaps automatically: the same packet appearing in two family images represents one element of $\mathfrak{M}_5^{\text{stand}}$.

Remark 3.2 (Representation exactness versus classification exactness). *Equation (2) is exact as an image representation once the domain cover and packet biconditional are established. An executed exact census requires more: explicit stratum domains, quotient parameters, packet values or packet moduli, failure boundaries, and overlap maps throughout the family images.*

3.4 Four levels of closure

We distinguish:

1. *candidate exactness*: every same-target source architecture has a PCR-3 presentation;
2. *survivor exactness*: every simultaneous survivor and only such survivors occur in A_5 ;
3. *identity exactness*: packet equality is exactly SCIC equality;
4. *classification exactness*: the full packet image has been decomposed into an explicit complete stratum atlas.

The theorem chain below establishes the first three. The fourth remains open for the general family images.

3.5 Fibre language

The packet map partitions A_5 into fibres

$$C_s := \{x \in A_5 : \Xi_5(x) = s\}, \quad s \in \mathfrak{M}_5^{\text{stand}}.$$

By packet completeness, these are exactly the SCIC classes. A source-provenance fibre can contain several family charts, while a single family chart can intersect many packet fibres.

Proposition 3.3 (Family ownership is secondary). *For every standing class $q \in \mathfrak{Q}_5^{\text{SCIC}}$, the set of family presentations above q is a provenance object. It does not refine q , and choosing one preferred family representative is unnecessary for quotient identity.*

Proof. All representatives in the fibre have the same packet and hence the same SCIC identity. Family ownership is retained in $\text{Prov}(q)$ for reconstruction, but the quotient element is the packet fibre itself. $\square$

3.6 Classification levels

Status	Condition
Exact finite census	The entire packet image is reduced to a finite explicit packet set.
Exact countable/moduli census	Every family image is decomposed into exact countable or parameterized packet strata with complete overlaps.
Exact packet-image representation	The quotient is canonically represented as the union of complete family packet images, but those images are not yet fully stratified.
Open representation problem	Domain coverage, packet completeness, or the packet-image representation itself remains unresolved.

The results proved here reach the third level and identify four explicit zero-dimensional packet strata within the representation.

4 Standing-faithful normal form and packet completeness

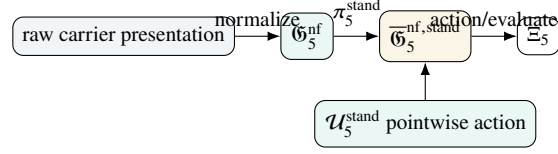


Figure 2: Normal-form, standing-descent, continuation-action, and packet architecture.

4.1 Primitive standing-generator system

PCR-5 defines $\mathcal{W}_5^{\text{stand}}$ as the lawful-redescription-invariant saturation of the terminal PCR-2 target uses, PCR-4 survival relations, and PCR-5 role/compatibility/failure uses under finite target-lawful products, restrictions, incidence tests, and fail-closed recodings. We make that saturation executable.

Let

$$\begin{aligned} \mathcal{G}_5 &= \mathcal{G}_T \sqcup \mathcal{G}_R \sqcup \mathcal{G}_D \sqcup \mathcal{G}_C \\ &\sqcup \mathcal{G}_M \sqcup \mathcal{G}_B \sqcup \mathcal{G}_F. \end{aligned}$$

be the primitive generator index family. A generator $g \in \mathcal{G}_X$ has a fixed codomain Y_g and is one of the following imported semantic primitives:

Type	Primitive source	Generator content
T	PCR-2 target-use primitives	charged-current records, event-complete channel values, visible response classes, scale/calibration and target-facing support values;
R	PCR-5 role/compatibility primitives	complete-role standing, eligibility, ownership, joint-incidence and common-cone role use;
D	PCR-5 no-repair deletion primitives	kindwise preservation/destruction/fail-closed response for every PCR-3 source kind admitted before terminal evaluation;
C	PCR-2/PCR-5 continuation execution primitives	support state and protocol/redescription data needed to act by the generators of $\mathcal{U}_5^{\text{stand}}$;
M	PCR-2/PCR-4 matching primitives	oriented source-to-effective matching, retained resolution, threshold/order and no-backflow status;
B	PCR-2/PCR-5 boundary primitives	source/detector attachment, boundary reference, ownership and lawful boundary transport;
F	PCR-4/PCR-5 failure primitives	same-target obstruction, undefinedness, deformation boundary, relation failure and fail-closed disposition.

Define the raw evaluation vector

$$\mathbb{G}_5^{\text{nf}}(x) := (g(x))_{g \in \mathcal{G}_5} \in \prod_{g \in \mathcal{G}_5} Y_g.$$

The seven types are finite; the index sets $\mathcal{G}_X$ and codomains Y_g can be symbolic, infinite, or categorical.

Theorem 4.1 (Census-generator factorization). *For every $w \in \mathcal{W}_5^{\text{stand}}$ there exists a target-lawful map*

$$F_w : \text{im } \mathfrak{G}_5^{\text{nf}} \longrightarrow Y_w$$

such that

$$w = F_w \circ \mathfrak{G}_5^{\text{nf}}.$$

Moreover the factorization is generated inductively from the primitive projections by exactly the closure operations used in PCR-5 to form $\mathcal{W}_5^{\text{stand}}$.

Proof. Let $\mathcal{A}(\mathcal{G}_5)$ be the smallest family of functionals on A_5 containing every primitive generator and closed under the finite operations used in the PCR-5 definition of $\mathcal{W}_5^{\text{stand}}$: target-lawful products, restrictions, incidence tests, and fail-closed recodings. For a primitive g , take F_g to be the corresponding coordinate projection of $\mathfrak{G}_5^{\text{nf}}$. If $w = (w_1, \dots, w_n)$ is a finite product and each $w_i = F_i \circ \mathfrak{G}_5^{\text{nf}}$, take $F_w = (F_1, \dots, F_n)$. If $w = \rho \circ w_0$ is a lawful restriction or fail-closed recoding, take $F_w = \rho \circ F_{w_0}$. If w is an incidence test formed from finitely many previously factored inputs, compose the corresponding incidence map with their product factorization. Thus every member of $\mathcal{A}(\mathcal{G}_5)$ factors through $\mathfrak{G}_5^{\text{nf}}$ by structural induction.

PCR-2 target-use factorization, PCR-4 survival-relation exhaustion, and PCR-5 role/compatibility/failure closure identify the primitive semantic layers entering $\mathcal{W}_5^{\text{stand}}$; no further same-target primitive layer is admitted by SCIC activation. Hence $\mathcal{W}_5^{\text{stand}} = \mathcal{A}(\mathcal{G}_5)$ at the fixed target. A proposed additional standing functional outside this algebra would therefore be either a new target/source coordinate or a direct falsifier of the imported semantic-exhaustion theorem. [2–4, 6] $\square$

4.2 Standing-faithful descent

Raw normal forms can retain lawful execution or source-history data invisible to terminal standing. For raw values G, H , define

$$G \equiv_{\text{stand}}^{\text{nf}} H \iff \text{Eval}_{5,w}^{\text{nf}}(G) = \text{Eval}_{5,w}^{\text{nf}}(H) \quad \forall w \in \mathcal{W}_5^{\text{stand}}.$$

Let

$$\overline{\mathfrak{G}}_5^{\text{nf,stand}} = \mathfrak{G}_5^{\text{nf}}(A_5) / \equiv_{\text{stand}}^{\text{nf}}$$

and π_5^{stand} be the quotient map.

Theorem 4.2 (Standing-faithful normal-form descent). *The quotient $\overline{\mathfrak{G}}_5^{\text{nf,stand}}$ removes every evaluator-invisible raw distinction and no evaluator-visible distinction. Unequal classes are separated by at least one admitted standing evaluator.*

Proof. This is the kernel quotient of the joint evaluator map

$$G \longmapsto (\text{Eval}_{5,w}^{\text{nf}}(G))_{w \in \mathcal{W}_5^{\text{stand}}}.$$

Kernel equality is precisely evaluator equality, so the first isomorphism theorem gives the claim. $\square$

4.3 Executable continuation action

PCR-5 fixes $\mathcal{U}_5^{\text{stand}}$ as the monoid generated by the identity, PCR-2 lawful protocol/support continuations, and identity-preserving source/interface redescrptions, all as total fail-closed same-target endomorphisms of A_5 . For a generator u of $\mathcal{U}_5^{\text{stand}}$, define its action on the primitive vector by

$$A_u(\mathfrak{G}_5^{\text{nf}}(x)) := \mathfrak{G}_5^{\text{nf}}(u(x)).$$

This is well defined on $\text{im } \mathfrak{G}_5^{\text{nf}}$: if $\mathfrak{G}_5^{\text{nf}}(x) = \mathfrak{G}_5^{\text{nf}}(y)$, then by theorem 4.1 every standing functional agrees on x, y ; for each primitive standing evaluator g , the continued functional $g \circ u$ is a licensed standing-

continuation evaluation and therefore factors through the same raw vector. The execution coordinates in $\mathcal{G}_C$ are transported by the imported PCR-2 continuation/redescription generator itself. Equality of those raw execution coordinates is preserved by each continuation generator, so the $\mathcal{G}_C$ component is representative-independent. Arbitrary $u \in \mathcal{U}_5^{\text{stand}}$ acts by composition of these generator actions.

The resulting coordinate rules are:

Type	Action under a continuation generator u
T	evaluate the same target-record/support primitive on $u(x)$; fixed target anchors are preserved and an internal undefined value is replaced by its PCR-5 fail-closed value;
R	transport role incidence, ownership, eligibility and common-cone use along the identity-preserving continuation; destruction of a constitutive path is recorded fail closed;
D	for each fixed source kind κ , replace $D_\kappa(x)$ by the same typed no-repair diagnostic $D_\kappa(u(x))$; the source kind and nondeleted data are not returned;
C	compose continuation execution data: the value at later labeled v is the raw execution data of $v(u(x))$;
M	transport only matching arrows that preserve retained resolution and source identity; a source-changing or resolution-changing arrow is not in $\mathcal{U}_5^{\text{stand}}$ and is recorded instead by matching/failure data;
B	transport the fixed source/detector boundary and ownership references along the same-target continuation; boundary mismatch is fail closed;
F	propagate the same-target failure domain and obstruction values under u , preserving undefinedness rather than repairing it.

Set $\text{Act}_5^{\text{nf}}(u, G) = A_u(G)$ for generators and extend by monoid composition. Define

$$\overline{\text{Act}}_5^{\text{stand}}(u, G) = \pi_5^{\text{stand}}(\text{Act}_5^{\text{nf}}(u, G)).$$

Theorem 4.3 (Standing-continuation action reconstruction). *The maps Act_5^{nf} form a total fail-closed action of $\mathcal{U}_5^{\text{stand}}$ on $\text{im } \mathfrak{G}_5^{\text{nf}}$. For every $w \in \mathcal{W}_5^{\text{stand}}$,*

$$\text{Eval}_{5,w}^{\text{nf}}(\text{Act}_5^{\text{nf}}(u, \mathfrak{G}_5^{\text{nf}}(x))) = w(u(x)).$$

Identity and composition satisfy

$$\text{Act}_5^{\text{nf}}(1, G) = G, \quad \text{Act}_5^{\text{nf}}(u_2 u_1, G) = \text{Act}_5^{\text{nf}}(u_2, \text{Act}_5^{\text{nf}}(u_1, G)).$$

Proof. For every generator of $\mathcal{U}_5^{\text{stand}}$, the coordinate table above is simply evaluation of the imported primitive semantic data on the same-target point $u(x)$; totality follows from PCR-5's fail-closed continuation typing. The factorization theorem applied to the continued standing use gives the evaluator identity. The monoid laws on A_5 then give identity and composition on the image of $\mathfrak{G}_5^{\text{nf}}$. Since π_5^{stand} is applied only after the raw action, evaluator-invisible transformed detail is removed without erasing latent execution information before continuation. $\square$

4.4 Complete packet

Define the pointwise continuation profile

$$C_5^{\text{act}}(x) : \mathcal{U}_5^{\text{stand}} \rightarrow \overline{\mathfrak{G}}_5^{\text{nf, stand}}, \quad u \mapsto \overline{\text{Act}}_5^{\text{stand}}(u, \mathfrak{G}_5^{\text{nf}}(x)).$$

Set

$$\Xi_5(x) = (\mathcal{T}_5^{\text{face}}(x), \mathcal{R}_5^{\text{stand}}(x), \Delta_5^{\text{del}}(x), C_5^{\text{act}}(x), \mathcal{M}_5^{\text{match}}(x), \mathcal{B}_5^{\text{bdy}}(x), \mathcal{F}_5^{\text{same}}(x)). \quad (3)$$

All non-continuation entries are standing-evaluator values or lawful finite recodings thereof.

Theorem 4.4 (Census-packet completeness). *For all $x, y \in A_5$,*

$$\boxed{\Xi_5(x) = \Xi_5(y) \iff \Sigma_5^{\text{SCIC}}(x) = \Sigma_5^{\text{SCIC}}(y).} \quad (4)$$

Consequently

$$\Omega_5^{\text{SCIC}} \cong \text{im } \Xi_5.$$

Proof. If the complete SCIC signatures agree, every non-continuation packet coordinate agrees because it is a standing-evaluator value or lawful recoding. For each same labeled u , every evaluator agrees on $u(x)$ and $u(y)$; hence the transformed raw vectors have equal π_5^{stand} -images and the pointwise continuation profiles agree.

Conversely, packet equality gives $C_5^{\text{act}}(x)(u) = C_5^{\text{act}}(y)(u)$ for each same labeled u . By standing-faithful descent every $w \in \mathcal{W}_5^{\text{stand}}$ has the same value on those classes, and by action reconstruction this common value is $w(u(x)) = w(u(y))$. Thus all entries of the complete SCIC signatures agree. $\square$

Corollary 4.5 (No packet over-resolution). *A difference in raw source description, family name, basis, presentation, or provenance cannot define a PCR-5A class unless it changes an admitted standing-evaluator value after some licensed continuation.*

4.5 First-difference exhaustion

Theorem 4.6 (Packet first-difference exhaustion). *If $x \not\sim_{\text{SCIC}} y$, then some admitted evaluator/continuation pair differs and therefore at least one of the seven packet channels differs. Conversely every packet-channel difference is witnessed by an admitted standing evaluator after a labeled continuation.*

Proof. SCIC inequality supplies (w, u) with unequal values. Factor w through the primitive generator algebra and choose a primitive factor on which the two continued raw vectors first differ; its type supplies the packet channel. The converse follows because every packet component is itself an evaluator image or a lawful recoding of evaluator images. $\square$

4.6 No-second-classifier consequence

Corollary 4.7 (No hidden packet refinement). *A proposed same-target carrier classifier not factoring through Ξ_5 is either standing-inert, target-changing, or a contradiction of SCIC target completeness.*

Proof. Any same-target standing classifier belongs to the activated standing family and therefore factors through the primitive algebra and the complete packet. If it does not, it lacks standing authority at this target or falsifies the imported completeness theorem. $\square$

5 Exact family cover and parameter descent

5.1 Family images

For $F \in \mathfrak{F}_5^{\text{surv}}$, define

$$\mathcal{I}_F = \Xi_5(\iota_F(\Theta_F^{\text{surv}})).$$

Let Γ_F denote the family parameter/reparameterization groupoid.

Theorem 5.1 (Parameter descent). *The packet descends to each lawful family quotient*

$$\Xi_F : \Theta_F^{\text{surv}} / \Gamma_F \rightarrow \text{im } \Xi_5.$$

A parameter labels standing identity exactly when varying it changes the complete packet after lawful redescription. Fibre-internal, skin, and target-changing parameters do not define additional classes in $\mathfrak{Q}_5^{\text{SCIC}}$.

Proof. Lawful family morphisms preserve the inherited carrier trace and all admitted standing evaluations, hence preserve Ξ_5 . The converse classification follows from theorem 4.4 and the fixed target-update boundary. $\square$

Theorem 5.2 (Exact family-image cover).

$$\text{im } \Xi_5 = \bigcup_{F \in \mathfrak{F}_5^{\text{surv}}} \mathcal{I}_F.$$

No standing class lies outside this union.

Proof. By lemma 2.2, every point of A_5 has a principal family presentation. Applying Ξ_5 gives the stated union. Pure BASE, TOP, EXT, BDY, and ENV contributions are already represented in the established overlaps or HYB survivor locus. $\square$

5.2 Cross-family overlap relation

For presentations $x \in \Theta_F^{\text{surv}}$, $y \in \Theta_G^{\text{surv}}$, write

$$x \approx_{\Xi} y \iff \Xi_5(\iota_F x) = \Xi_5(\iota_G y).$$

By packet completeness this is the pullback of SCIC equality.

Theorem 5.3 (Representative-free exact quotient). *The quotient*

$$Q_5^{\text{exact}} := \left(\bigsqcup_{F \in \mathfrak{F}_5^{\text{surv}}} \Theta_F^{\text{surv}} / \Gamma_F \right) / \approx_{\Xi}$$

is canonically isomorphic to $\mathfrak{Q}_5^{\text{SCIC}}$.

Proof. The map sending a family presentation to its packet is surjective by theorem 5.2. Its fibres are exactly $\approx_{\Xi}$, which is SCIC equality by theorem 4.4. Quotienting by the fibres yields the image and hence $\mathfrak{Q}_5^{\text{SCIC}}$. $\square$

5.3 Exact family parameter groupoids

The lawful groupoid Γ_F includes basis, gauge, representation-chart, source-presentation, parameter-recharting, and duplicate-family transformations that preserve the complete active carrier trace. It does not include a map that changes source identity, target fibre, or a standing packet coordinate.

Lemma 5.4 (Family quotient soundness). *If θ and θ' lie in one Γ_F -orbit, then*

$$\Xi_5(\iota_F \theta) = \Xi_5(\iota_F \theta').$$

If the packets differ, no lawful reparameterization can identify the points.

Proof. The first statement follows from representative independence of ι_F and the packet. The second is witness soundness. $\square$

5.4 Overlap coverage beyond pairwise tables

A pairwise overlap matrix is useful exposition but not the proof of global exactness. Multi-family overlaps can be nontrivial, particularly in HYB. The common packet codomain handles arbitrary finite overlaps automatically: all presentations with one packet define one point, regardless of how many family charts contain it.

Theorem 5.5 (Finite-overlap coherence). *For any finite collection of family presentations $x_i \in \Theta_{F_i}^{\text{surv}}$, the following are equivalent:*

1. *all $\iota_{F_i}(x_i)$ lie in one SCIC class;*
2. *all packets $\Xi_5(\iota_{F_i}(x_i))$ are equal;*
3. *the presentations define one point of Q_5^{exact} .*

Proof. Pairwise packet equality is transitive and equivalent to SCIC equality. The quotient defining Q_5^{exact} is the equivalence closure of the same relation. $\square$

5.5 Why a finite family list is compatible with infinite classes

The six principal families classify source-construction form, not the number of instances. Each family can contain infinitely many symbolic templates and parameter strata. PCR-3 completeness places every finite source-coherent architecture in one family presentation or HYB; PCR-5A then counts standing packet values, not family labels.

6 Principal family packet images and unresolved stratification

For each principal family F , the PCR-4 survivor domain fixed by the inherited theorem chain and the complete packet define the image

$$\mathcal{I}_F = \{\Xi_5(\iota_F(\theta)) : \theta \in \Theta_F^{\text{surv}}\}.$$

This is an exact image *representation*. The purpose of this section is to separate the exact image representation from the additional stratum classification still required within those images.

6.1 LAW

Theorem 6.1 (LAW packet-image representation). *Every LAW survivor is represented by the seven-component packet restricted to the exact PCR-4 LAW domain. Two LAW points define one standing class exactly when their complete packets agree.*

Proof. PCR-4 equation (51) supplies the exact domain, PCR-3 supplies the source-normal-form groupoid, and theorem 4.4 supplies the exact fibre relation. $\square$

The Weinberg point x_W is one class in this image. The inherited theorem chain does not prove that every standing-visible nonlocal-memory, operator-order, matching, boundary, or failure variation either collapses to s_W or lies in a specifically enumerated alternative LAW stratum. Thus the internal LAW stratum atlas remains open.

6.2 LOCAL and HID

Theorem 6.2 (LOCAL/HID packet-image representation). *The exact PCR-4 LOCAL and HID survivor domains map into $\mathcal{I}_{\text{LOCAL}} \cup \mathcal{I}_{\text{HID}}$ by the complete packet, and packet equality is exactly SCIC equality on these domains.*

Proof. Apply PCR-4 equations (52) and (57), PCR-3 local/hidden normal forms, and theorem 4.4. $\square$

The full-rank three-singlet point x_{3N} represents s_{3N} . Multiplicity, active rank, hidden dimension and dilation size can refine the packet only when standing-visible. The inherited theorem chain does not prove a complete partition of these symbolic domains by those standing-visible combinations, so their executed stratum atlas remains open.

6.3 COMP and SOL

Theorem 6.3 (COMP/SOL packet-image representation). *The exact packet images are*

$$\mathcal{I}_{\text{COMP}} = \Xi_5(\iota_{\text{COMP}}(\Theta_{\text{COMP}}^{\text{surv}})), \quad \mathcal{I}_{\text{SOL}} = \Xi_5(\iota_{\text{SOL}}(\Theta_{\text{SOL}}^{\text{surv}})),$$

with fibres exactly equal to the restrictions of SCIC classes.

Proof. PCR-4 gives the exact implicit survivor domains and theorem 4.4 supplies the fibre equivalence. Constituent, derivation, persistence, topology, boundary and failure data survive precisely when they change an admitted packet coordinate. $\square$

No complete parameter-stratum classification of either image follows from the inherited results used here.

6.4 HYB

The HYB image is represented exactly by the survivor-restricted decorated category of section 7. Section 8 constructs two explicit non-anchor singleton strata in that image. Those calibrations do not classify every possible higher-arity or parameterized HYB packet.

6.5 What has been classified and what remains open

The distinction between established image structure and the remaining classification problem is summarized below:

Family	Established here	Remaining classification problem
LAW	exact domain, packet map, Weinberg anchor	partition the full symbolic LAW domain into packet-constant strata and prove all overlap maps;
LOCAL	exact domain, packet map, active benchmark	classify standing-visible rank/multiplicity/threshold strata;
HID	exact domain, packet map, hidden/local overlap at benchmark	classify minimal dilation/support/projection packet strata;
COMP	exact domain and packet-fibre relation	classify constituent/derivation packet strata and overlaps;
SOL	exact domain and packet-fibre relation	classify persistence/topology/endability packet strata and overlaps;
HYB	survivor category, packet map, two mixed singleton strata	classify every remaining decorated-hybrid packet stratum, including higher-arity cross effects.

Theorem 6.4 (Family-stratification boundary). *The data proved above are sufficient for the exact packet-image representation theorem but insufficient for an executed exact stratum census in the sense of Definition 2 of section 1. In particular, the symbolic sets $\mathcal{I}_{\mathbb{F}}$ are not replaced here by a complete list of packet-constant subdomains or parameterized packet moduli.*

Proof. The exact domains and packet maps determine the images set-theoretically, but the displayed family results deliberately leave open whether standing-visible continuation, matching, boundary, failure, rank, constituent, persistence, or higher-arity coordinates generate further strata. An executed census requires those alternatives to be dispositioned over the full symbolic domains; no theorem in the inherited chain performs that additional reduction. $\square$

Proposition 6.5 (Weinberg anchor is not a family definition). *A LAW survivor belongs to C_W exactly when its complete packet equals s_W ; absence of an active new primitive node is not sufficient.*

Proof. The no-new-node condition fixes only one source aspect. SCIC identity requires all packet coordinates to agree. $\square$

Proposition 6.6 (No generation-cardinality shortcut). *The three generations of the fixed Standard Model do not force three right-handed neutrinos, and the explicit 3N benchmark does not identify every LOCAL or HID survivor with C_{3N} .*

Proof. The generation theorem governs the fixed Standard-Model carrier. PCR-1 leaves neutrino completion coordinates open, and SCIC identity is packet equality rather than cardinality matching. $\square$

Proposition 6.7 (No effective-field backflow). *Equality of a COMP, SOL, HID, or HYB effective output with a LAW or LOCAL coefficient does not identify the carriers unless the complete packets agree.*

Proof. Oriented matching can forget constituent, persistence, hidden-support, boundary and failure data. The packet theorem preserves every standing-visible difference and collapses only evaluator-invisible structure. $\square$

7 Survivor-restricted decorated hybrids

7.1 Pre-survivor decorated category

Let $\text{HybDec}_5^{\text{pre}}$ be the groupoid of finite PCR-3 source-coherent decorated hybrid hypergraphs. Vertices carry source subdiagrams; hyperedges carry the actual source incidence, common-point/common-cone, anomaly, vacuum, matching, continuation, boundary, and failure data. Hyperedges may have any finite arity.

The exact PCR-4 evaluator gives each pre-survivor object one disposition:

Survivor, SameTargetFailure, Target/SourceUpdate.

Only the first enters A_5 .

Definition 7.1 (Survivor decorated category). *Let*

$$\text{HybDec}_5^{\text{surv}} := \text{HybDec}_5^{\text{pre}} \times_{\Theta_{\text{HYB}}^{\text{part}}} \Theta_{\text{HYB}}^{\text{surv}}.$$

For $(H, \gamma_H) \in \text{HybDec}_5^{\text{surv}}$, define

$$\iota_{\text{HYB}}^{\text{dec}}(H, \gamma_H) := \iota_{\text{HYB}}(\rho_{\text{HYB}}(H, \gamma_H)) \in A_5.$$

Theorem 7.2 (No PCR-3-to-PCR-4 promotion). *The map $\iota_{\text{HYB}}^{\text{dec}}$ is defined exactly on the survivor pullback. A decorated candidate carrying a PCR-4 failure or target/source update has no image in A_5 and no standing-census class.*

Proof. The pullback definition enforces PCR-4 membership before the map to the survivor quotient. PCR-3 source coherence supplies candidate formation only; PCR-4 theorem 18.1 supplies survival. $\square$

7.2 Gluing-indexed composition

For two compatible decorated subpresentations P, Q , let $\text{Glue}_5(P, Q)$ be the groupoid of lawful gluing objects. For $\gamma \in \text{Glue}_5(P, Q)$, write

$$P \boxplus_{5,\gamma} Q = NF_{\text{hyb}}^{\text{pre}}(P \sqcup_{\gamma} Q).$$

There is no gluing-independent packet operation: distinct gluings of the same constituent packets can produce different standing cross effects.

Theorem 7.3 (Decorated-hybrid colimit). *There is a standing-faithful map*

$$\text{Colim}_5^{\text{hyb}} : \text{HybDec}_5^{\text{surv}} \rightarrow \overline{\mathfrak{G}}_5^{\text{nf,stand}}$$

obtained by taking the complete hybrid normal form and applying π_5^{stand} . It retains every standing-bearing higher-arity hyperedge. Different binary decomposition trees of the same complete decorated hybrid have the same standing-faithful colimit.

Proof. PCR-3 finite-hybrid closure and incidence reconstruction give the complete decorated hypergraph independently of a binary decomposition. PCR-4 evaluates the complete architecture rather than constituent verdicts. Lawful decomposition changes are isomorphisms of the same decorated object; raw normal forms may differ by presentation, but their standing-evaluator quotient agrees. $\square$

Corollary 7.4 (Packet decomposition independence). *For any two lawful decomposition trees τ_1, τ_2 of the same surviving decorated hybrid,*

$$\Xi_5^{(\tau_1)}(H, \gamma_H) = \Xi_5^{(\tau_2)}(H, \gamma_H).$$

Proof. The common decorated object determines the same standing colimit, continuation action, and seven packet components. Apply hybrid packet reconstruction and theorem 4.4. $\square$

Theorem 7.5 (Exact HYB image).

$$\mathcal{I}_{\text{HYB}} = \{ \Xi_5(\iota_{\text{HYB}}^{\text{dec}}(H, \gamma_H)) : (H, \gamma_H) \in \text{HybDec}_5^{\text{surv}} \}.$$

This image includes all surviving topological, extended, boundary, environmental, composite, solitonic, local, hidden, and law-bearing cross architectures.

7.3 Pre-survivor evaluation functor

The distinction between source coherence and simultaneous survival can be expressed by a disposition functor

$$\text{Disp}_4^{\text{HYB}} : \text{HybDec}_5^{\text{pre}} \longrightarrow \{\text{Survivor}, \text{SameTargetFailure}, \text{Target/SourceUpdate}\}.$$

A source-coherent decorated hypergraph can fail through any intrinsic or cross-signature coordinate. The survivor pullback $\text{HybDec}_5^{\text{surv}}$ is the full subgroupoid on the first fibre.

Lemma 7.6 (Failure preservation). *A decorated hybrid in a failure or target-update fibre cannot be moved into $\text{HybDec}_5^{\text{surv}}$ by changing its binary decomposition tree, choosing another family owner, or replacing a failed cross relation with a constituent pass result.*

Proof. The disposition is evaluated on the complete decorated hypergraph and all cross relations. A decomposition is presentation; it does not alter the underlying source equations or PCR-4 point. $\square$

7.4 Higher-arity cross effects

The complete decoration can carry a k -ary relation that has no pairwise reduction: a common-cone constraint, multi-sector anomaly cancellation, a junction/fusion law, or a shared-parameter condition. The hybrid colimit therefore retains hyperedges directly. Binary gluing notation is only a method for constructing a presentation of a fixed decorated object.

Theorem 7.7 (Higher-arity completeness). *Every standing-bearing finite-arity hybrid relation occurs in the complete decorated invariant and is evaluated in a corresponding packet component. No pairwise-factorization assumption enters $\mathcal{I}_{\text{HYB}}$.*

Proof. PCR-3's finite coherent-hypergraph presentation includes all incidence hyperedges and coherence cells. PCR-4's hybrid map evaluates every cross-signature residual created by the gluing. The standing-faithful packet retains exactly those evaluated effects visible to $\mathcal{W}_5^{\text{stand}}$. $\square$

7.5 Decorated reconstruction versus constituent verdicts

Even if every constituent packet equals an anchor, the complete hybrid can be third because of a new cross effect. Conversely, a constituent distinction can become standing-inert inside the complete hybrid. Thus the correct computation is

$$(H, \gamma_H) \mapsto \iota_{\text{HYB}}^{\text{dec}}(H, \gamma_H) \mapsto \Xi_5,$$

not a product of constituent class labels.

Corollary 7.8 (No constituentwise endpoint rule). *There is no lawful rule*

$$\{\text{constituent anchor labels}\} \longrightarrow \text{hybrid standing class}$$

independent of the complete decorated gluing.

Proof. Different gluing objects between identical constituent packets can have unequal cross effects. Packet equality is evaluated only after the complete surviving architecture is formed. $\square$

8 Mixed law-plus-singlet calibration

8.1 Source algebra and explicit HYB parameter objects

Let $\kappa \in \text{Sym}_3(\mathbb{C})$ denote the low-energy Majorana coefficient, with conventional overall factors suppressed. A mixed source architecture contains an independent LAW source κ_W and an active singlet contribution

$$\kappa_N = Y_\nu M_R^{-1} Y_\nu^T, \quad \kappa_{\text{tot}} = \kappa_W + \kappa_N.$$

The matched singlet output κ_N is not retyped as an independent LAW source.

Lemma 8.1 (Takagi singlet realization). *For every $K = UDU^T \in \text{Sym}_3(\mathbb{C})$ and every positive diagonal M_R ,*

$$Y = UD^{1/2} M_R^{1/2}$$

satisfies $YM_R^{-1} Y^T = K$. Multiplication of Y by a scalar phase $e^{i\alpha}$ multiplies the induced symmetric coefficient by $e^{2i\alpha}$.

Proof. Direct multiplication gives $UD^{1/2} M_R^{1/2} M_R^{-1} M_R^{1/2} D^{1/2} U^T = UDU^T$. The phase statement follows because the transpose, not adjoint, appears in the symmetric coefficient. $\square$

For the two mixed witnesses we now construct complete points of the PCR-4 HYB pre-survivor domain. The coordinate package used below is exactly the simultaneous data listed by the imported PCR-4 family evaluator: fixed target/boundary records; source hypergraph and one parameter point; event-complete channel

and response data; support/continuation; gauge/anomaly data; one vacuum/deformation branch; oriented matching; activated-extension status; all intrinsic and cross relation residuals; and a common cone. We denote such a point by

$$\theta_{\text{HYB}} = (\mathcal{T}_*, H, \lambda, \mathcal{E}, \mathcal{S}, \mathcal{A}, \mathcal{V}, \mathcal{M}, \mathcal{X}, \mathcal{R}, C).$$

This is notation for the corresponding source-interpreted PCR-4 parameter object, not a new survivor predicate.

HYB import boundary. The mixed proofs use only the PCR-4 HYB facts fixed by the inherited theorem chain: the complete family evaluator, the inverse-image definition of the HYB survivor locus, and the rule that its input is the complete decorated hybrid point. No separate “constituentwise-survival” theorem and no additional PCR-4 theorem number is imported. Thus the synergistic construction is judged by the displayed complete-point residual/cone calculation below, not by an assertion about whether either rank-one constituent survives on its own.

8.2 Anchor deletion pairs

Let κ_{law} and κ_{newloc} be the independent LAW and new-local source kinds. Define

$$\mathbf{d}_{W \oplus 3N}(x) = (D_{\kappa_{\text{law}}}^{\text{role}}(x), D_{\kappa_{\text{newloc}}}^{\text{role}}(x)).$$

Theorem 8.2 (Anchor deletion-pair theorem).

$$\mathbf{d}_{W \oplus 3N}(x_W) = (0, 1), \quad \mathbf{d}_{W \oplus 3N}(x_{3N}) = (1, 0).$$

Proof. For x_W , deleting the active LAW source leaves the strict massless-neutrino BASE architecture, which PCR-4 excludes from the required role packet; new-local deletion is a typed no-op because no such source is active. For x_{3N} , there is no independent LAW source and matching backflow cannot turn the derived EFT coefficient into one; LAW deletion is therefore a typed no-op, whereas new-local deletion removes the constitutive singlet path and destroys the complete role packet by PCR-5. $\square$

8.3 Phase-redundant mixed point

Choose a full target-realizing symmetric coefficient

$$K_* = UDU^T$$

with the target-required response-mode separation and nonalignment. Put $\omega = e^{2\pi i/3}$, choose a positive diagonal M_R , and define

$$\kappa_W^{\text{red}} = K_*, \quad Y_{\text{red}} = e^{i\pi/3} U D^{1/2} M_R^{1/2}, \quad \kappa_N^{\text{red}} = Y_{\text{red}} M_R^{-1} Y_{\text{red}}^T = \omega K_*.$$

Then

$$\kappa_{\text{tot}}^{\text{red}} = (1 + \omega) K_* = e^{i\pi/3} K_*, \quad \kappa_{\text{tot}}^{\text{red}} (\kappa_{\text{tot}}^{\text{red}})^\dagger = K_* K_*^\dagger.$$

Define $\theta_{\text{red}} \in \Theta_{\text{HYB}}^{\text{part}}$ by the following one-point data:

1. $\mathcal{T}_*$: the same fixed charged-current preparation/detection records and target instrument as the law-only benchmark;
2. H_{red} : the Standard-Model lepton/Higgs support carrying both the independent $O_5(K_*)$ LAW hyperedge and the active singlet nodes $(N_R, Y_{\text{red}}, M_R)$;
3. λ_{red} : one common electroweak vacuum and one common parameter point containing K_*, Y_{red}, M_R ;
4. $\mathcal{E}_{\text{red}}$: the event-complete channel generated by $\kappa_{\text{tot}}^{\text{red}} (\kappa_{\text{tot}}^{\text{red}})^\dagger$, with the same null-outcome completion as the benchmark;

5. $\mathcal{S}_{\text{red}}$: the benchmark source-to-detector causal/support continuation, now sourced by the complete mixed hypergraph;
6. $\mathcal{A}_{\text{red}}$: Standard-Model gauge data plus gauge-singlet N_R ; the LAW hyperedge is gauge invariant and the singlet contribution adds no perturbative gauge anomaly;
7. $\mathcal{V}_{\text{red}}$: the single electroweak vacuum/deformation branch shared by both terms;
8. $\mathcal{M}_{\text{red}}$: the oriented matching arrow $(Y_{\text{red}}, M_R) \mapsto \kappa_N^{\text{red}}$, with the independent LAW source left unmapped and unmerged;
9. $\mathcal{X}_{\text{red}}$: no additional TOP/EXT/BDY/ENV source is activated; every optional extension coordinate therefore takes the imported inactive value;
10. $\mathcal{R}_{\text{red}}$: every intrinsic and cross residual of the complete gluing is zero: cross anomaly is zero, both source terms use the same vacuum so cross-vacuum mismatch is zero, the singlet matching equation holds exactly, the source sum equals $\kappa_{\text{tot}}^{\text{red}}$, and there is no nontrivial additional higher-arity relation beyond the shared support/parameter incidence already encoded in the hypergraph;
11. $\mathcal{C}_{\text{red}}$: the cone with apex the complete mixed source action and legs given by restriction to the LAW subdiagram, restriction to the singlet subdiagram, the effective map sending the pair to $\kappa_W^{\text{red}} + \kappa_N^{\text{red}}$, and the fixed target evaluation. The two routes from the apex to the target commute because target evaluation is performed on that same summed coefficient and common boundary data.

Theorem 8.3 (Redundant mixed-survivor theorem). *The point θ_{red} lies in $\Theta_{\text{HYB}}^{\text{surv}}$. Its image $x_{\text{red}} = \iota_{\text{HYB}}(\theta_{\text{red}}) \in A_5$ satisfies*

$$\mathbf{d}_{W \oplus 3N}(x_{\text{red}}) = (1, 1).$$

Proof. PCR-4 survivor membership is $\widehat{\Phi}_{\text{HYB}}(\theta) \in \mathcal{Z}_4^{\text{sem}}$, evaluated on the complete hybrid point, not by multiplying constituent verdicts. We verify the full coordinate list on θ_{red} . Target records and the event-complete channel agree with the benchmark because the charged-current boundary is fixed and the total Hermitian propagation kernel equals $K_* K_*^\dagger$. The response-mode and nonalignment coordinates therefore agree. The support/continuation coordinate is the benchmark continuation on the same lepton/Higgs and boundary support. Gauge and anomaly coordinates pass because the LAW operator is gauge invariant and the added neutrinos are gauge singlets. Vacuum/deformation uses one common branch. Matching passes by the displayed exact equation for $Y_{\text{red}} M_R^{-1} Y_{\text{red}}^T$, with no backflow into the independent LAW source. No optional extension source is active. Every cross residual listed above is zero at the same parameter point, and $\mathcal{C}_{\text{red}}$ explicitly supplies a nonempty common cone. Hence $\widehat{\Phi}_{\text{HYB}}(\theta_{\text{red}}) \in \mathcal{Z}_4^{\text{sem}}$.

Deleting the LAW source holds Y_{red}, M_R and all nondeleted data fixed and leaves ωK_* , whose Hermitian square is $K_* K_*^\dagger$; the complete role packet remains standing. Deleting the singlet source leaves K_* , the law-only benchmark coefficient. Both typed no-repair deletion responses are therefore one. $\square$

8.4 Rank-split synergistic mixed point

Choose

$$K_* = U \text{diag}(d_1, d_2, 0) U^T = d_1 u_1 u_1^T + d_2 u_2 u_2^T, \quad d_1 \neq d_2 > 0,$$

with U non-diagonal in the charged-current basis and d_1^2, d_2^2 chosen to realize the two target phase-frequency intervals relative to the zero mode. Let

$$\kappa_W^{\text{syn}} = d_1 u_1 u_1^T, \quad M > 0, \quad Y_{\text{syn}} = \sqrt{d_2 M} u_2, \quad \kappa_N^{\text{syn}} = Y_{\text{syn}} M^{-1} Y_{\text{syn}}^T = d_2 u_2 u_2^T.$$

Then $\kappa_{\text{tot}}^{\text{syn}} = K_*$.

Define $\theta_{\text{syn}} \in \Theta_{\text{HYB}}^{\text{part}}$ by the same target, boundary, support and electroweak vacuum data as above, with the LAW rank-one hyperedge $d_1 u_1 u_1^T$, one active gauge-singlet node (N_R, Y_{syn}, M) , the exact matching

equation for κ_N^{syn} , no optional extension source, and the common cone whose effective leg sends the two source terms to their sum K_* . All cross anomaly, cross vacuum, matching and global relation residuals are zero for the same reasons as in the redundant point.

Theorem 8.4 (Synergistic mixed-survivor theorem). *The point θ_{syn} lies in $\Theta_{\text{HYB}}^{\text{surv}}$. Its image $x_{\text{syn}} \in A_5$ satisfies*

$$\mathbf{d}_{W \oplus 3N}(x_{\text{syn}}) = (0, 0).$$

Proof. The complete hybrid coefficient is exactly the target-realizing K_* . Hence the fixed boundary records, event-complete channel, two response-mode classes, and nonalignment coordinates pass at the one common point. Support/continuation, gauge/anomaly, vacuum/deformation, matching, extension status and the common cone are the explicitly constructed data above, and every cross residual vanishes. Thus the complete PCR-4 hybrid evaluator is in $\mathcal{Z}_4^{\text{sem}}$ and θ_{syn} is a survivor.

No separate constituent-survival premise is used. This follows directly from the imported hybrid survivor definition: the predicate is applied to $\widehat{\Phi}_{\text{HYB}}(\theta_{\text{syn}})$ for the complete decorated point. After LAW deletion the remaining singlet coefficient has rank one and only one nonzero singular value; after new-local deletion the remaining LAW coefficient likewise has rank one. Each deleted architecture therefore has at most one nontrivial phase-frequency scale and fails the target requirement of two nuisance-separated visible response-mode classes. The no-repair values are both zero. $\square$

8.5 Four standing classes

Theorem 8.5 (Four deletion packets are SCIC-distinct). *The four survivors $x_W, x_{3N}, x_{\text{red}}, x_{\text{syn}}$ are pairwise SCIC-inequivalent.*

Proof. Their deletion pairs are the four distinct elements of $\{0, 1\}^2$. PCR-5 proves every defined coordinate of the kindwise deletion family is a same-target standing-bearing failure/continuation consequence. The identity continuation therefore exposes each pair inside the complete SCIC signature. Unequal pairs imply unequal signatures. $\square$

Corollary 8.6 (Exact-two and exact-three exclusion).

$$|\mathcal{Q}_5^{\text{SCIC}}| \geq 4.$$

In particular, exact two and exact three are false at the fixed target.

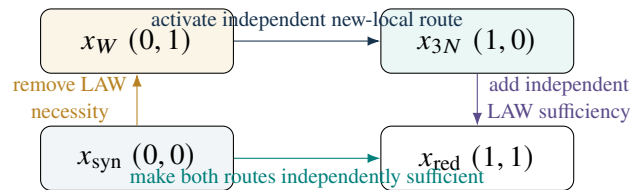


Figure 3: The four certified law/new-local deletion pairs. The square is a lower-bound witness, not a complete drawing of the remaining family packet images.

8.6 Deletion is evaluated without retuning

The deletion response holds the parameter point and all nondeleted source data fixed. It does not retune a coefficient, promote a matched output to an independent source, change vacuum branch, or substitute another family presentation.

Proposition 8.7 (Pairwise separation is independent of the unresolved family atlas). *The four deletion pairs are fixed on four explicit PCR-4 survivors before any global family stratification is attempted. Additional*

LAW, LOCAL, HID, COMP, SOL, or HYB classes cannot merge these four points without contradicting the standing deletion coordinates.

Proof. Every pair is evaluated on one fixed survivor with the already activated no-repair diagnostic. Other survivor points do not alter that evaluation. $\square$

9 Cross-family equality and overlap representation

9.1 Packet equality overrides provenance

For family presentations $x \in \Theta_F^{\text{surv}}$ and $y \in \Theta_G^{\text{surv}}$, define

$$x \approx_{\Xi} y \iff \Xi_5(\iota_F x) = \Xi_5(\iota_G y).$$

Theorem 9.1 (Cross-family identity rule).

$$[x]_{\text{SCIC}} = [y]_{\text{SCIC}} \iff x \approx_{\Xi} y.$$

Proof. Apply theorem 4.4 to the two images in A_5 . $\square$

Thus family ownership is provenance. Shared source components do not merge unequal packets, and distinct family labels do not separate equal packets.

9.2 Overlap representation

The set-theoretic overlap of two family packet images is exactly

$$\mathcal{I}_F \cap \mathcal{I}_G = \{s : \exists x, y \text{ with } \Xi_5(\iota_F x) = s = \Xi_5(\iota_G y)\}.$$

This formula is complete as an equality rule. It is not a classification of the overlap into anchor, discrete-third, or parameterized strata.

Proposition 9.2 (No double counting). *If a packet occurs in several family images, it contributes one element to $\mathfrak{M}_5^{\text{stand}}$. If two packets differ on an admitted standing coordinate, no family overlap can merge them.*

Proof. The first statement is ordinary union in the common packet codomain; the second is witness soundness of SCIC. $\square$

9.3 Known calibrated fibres

The following provenance inclusions are established explicitly:

- q_W has a LAW presentation;
- q_{3N} has LOCAL/HID presentations;
- q_{red} has the explicit HYB presentation θ_{red} ;
- q_{syn} has the explicit HYB presentation θ_{syn} .

Additional family presentations enter any one of these fibres only by complete packet equality.

9.4 Unresolved overlap atlas

A complete executed census must calculate every nonempty intersection $\mathcal{I}_F \cap \mathcal{I}_G$, and every higher multi-family intersection, as exact packet strata with presentation maps. The equivalence relation governing those intersections is fixed here, but their full stratum decomposition is not yet derived.

Theorem 9.3 (Overlap-classification boundary). *The packet equality rule is sufficient for the exact packet-image representation theorem, but a complete overlap atlas remains part of the open family-stratification burden.*

Proof. The representation theorem requires only that equal packet values be identified and that every family image be included. An executed census requires the stronger result of describing the intersections themselves by exact domains and packet strata. No such complete six-family overlap decomposition follows from the inherited results used here. $\square$

9.5 Anchor limits

A mixed stratum can approach an anchor when one source becomes inactive. PCR-3 activity reduction removes a genuinely null source before carrier identity is formed. A merely small but active source remains part of the candidate and must be evaluated; no coupling threshold is used to erase it.

10 Global packet-image representation

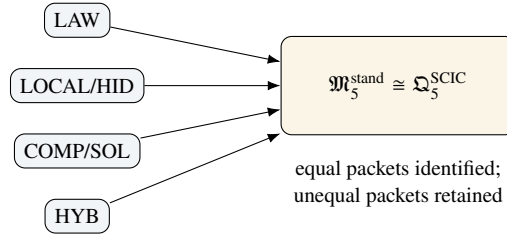


Figure 4: Family packet-image colimit in the common standing-packet codomain.

10.1 The packet-image union

Define

$$\mathfrak{M}_5^{\text{stand}} := \bigcup_{F \in \mathfrak{F}_5^{\text{surv}}} \mathcal{I}_F.$$

Theorem 10.1 (Global packet-image representation). *The map*

$$\Phi_5 : \mathfrak{Q}_5^{\text{SCIC}} \longrightarrow \mathfrak{M}_5^{\text{stand}}, \quad [x]_{\text{SCIC}} \longmapsto \Xi_5(x)$$

is a canonical bijection.

Proof. Well-definedness and injectivity follow from theorem 4.4. Surjectivity follows from the inherited family cover and the definition of the six packet images. No presentation choice enters the map. $\square$

Corollary 10.2 (Exact representation closure).

$$\mathfrak{Q}_5^{\text{SCIC}} \cong \mathfrak{M}_5^{\text{stand}}.$$

Every SCIC class is represented by one complete packet value from a PCR-4 survivor family, and every equal packet across family charts is identified exactly once.

10.2 What the representation theorem does not prove

The bijection does not itself compute the internal structure of the six images. In particular, the notation

$$\mathcal{I}_F = \text{im}(\Xi_5 \circ \iota_F)$$

is a mathematically exact image definition, but an executed census additionally requires the complete stratum decomposition demanded in theorem 6.4.

Let

$$Q_{\text{cal}} = \{s_W, s_{3N}, s_{\text{red}}, s_{\text{syn}}\} \subseteq \mathfrak{M}_5^{\text{stand}}$$

and define the unstratified remainder

$$\mathfrak{M}_5^{\text{rem}} := \mathfrak{M}_5^{\text{stand}} \setminus Q_{\text{cal}}.$$

Theorem 10.3 (Calibrated four-class subobject).

$$\mathfrak{M}_5^{\text{stand}} = Q_{\text{cal}} \sqcup \mathfrak{M}_5^{\text{rem}}, \quad |Q_{\text{cal}}| = 4.$$

Proof. The four packets are pairwise distinct by theorem 8.5; take their set complement in the exact packet image. $\square$

The remainder is defined but not yet stratified

The set $\mathfrak{M}_5^{\text{rem}}$ is an exactly defined image complement, not an already classified moduli space. Its complete family-stratum decomposition remains open, and the set-theoretic definition of the complement is not used as a substitute for that classification.

10.3 Equivalent quotient representation

Let

$$Q_5^{\text{rep}} := \left(\bigsqcup_{F \in \mathfrak{F}_5^{\text{surv}}} \Theta_F^{\text{surv}} / \Gamma_F \right) / \approx_{\Xi},$$

where $\theta \approx_{\Xi} \theta'$ iff their complete packets agree after the family presentation maps. Then

$$Q_5^{\text{rep}} \cong \mathfrak{Q}_5^{\text{SCIC}}.$$

This is a representative-free quotient presentation. It remains distinct from an executed atlas of the quotient's strata.

Theorem 10.4 (Representation upper-closure). *No additional same-target SCIC class can lie outside $\mathfrak{M}_5^{\text{stand}}$ without falsifying at least one fixed upstream input: PCR-3 candidate completeness, PCR-4 survivor completeness, SCIC activation, or packet completeness.*

Proof. Every $x \in A_5$ has a principal family presentation. Applying the complete packet places its SCIC class in $\mathfrak{M}_5^{\text{stand}}$. A putative further class outside the image would therefore require a survivor outside A_5 , a family-cover omission, a standing evaluator omitted by SCIC activation, or an incorrect packet biconditional. $\square$

10.4 Remaining PCR-5A classification problem

The unresolved problem is now precise:

Classify every $\mathcal{I}_F$ into exact packet-constant or packet-moduli strata and close every overlap.

The four-class theorem proves that any such final atlas contains at least four distinct classes; it does not determine how many additional discrete or parameterized strata occur.

11 Cardinality consequences and remaining classification

11.1 Exact two and exact three are excluded

Theorem 11.1 (No exact-two or exact-three endpoint). *Within the fixed PCR-5 target and comparison class,*

$$|\mathfrak{Q}_5^{\text{SCIC}}| \geq 4.$$

Hence $\mathfrak{Q}_5^{\text{SCIC}}$ is neither the two-anchor set nor any three-class extension of it.

Proof. Apply corollary 8.6. □

11.2 Established PCR-5A result

Theorem 11.2 (Bounded PCR-5A endpoint). *The results proved here give*

$$\mathfrak{Q}_5^{\text{SCIC}} \cong \mathfrak{M}_5^{\text{stand}}, \quad |\mathfrak{Q}_5^{\text{SCIC}}| \geq 4,$$

where $\mathfrak{M}_5^{\text{stand}}$ is the exact packet-image representation of the quotient. They do not prove that the general family images have been completely stratified.

Proof. Combine theorems 6.4 and 10.1 and corollary 8.6. □

Strongest interpretation

One terminal complete-carrier role admits at least four irreducible same-target standing-carrier classes. The entire SCIC quotient has a faithful packet-image representation, while its complete discrete, continuous, or categorical stratum atlas remains to be determined.

11.3 No realization selection

The strengthened lower bound does not alter PCR-5's realization verdict. Multiple standing classes remain and the target provides neither a singleton-incidence trace nor a theorem eliminating every rival but one. Therefore

$$\text{IrreducibleStructuralUnderdetermination}_{\text{car}}$$

remains the realization-level endpoint.

11.4 What is numerically determined

The lower bound four is established; exact four is not. A different continuation action, matching threshold, constituent relation, persistent profile, boundary incidence, or failure domain can define an additional packet class whenever that difference is standing-visible. No finite cardinality beyond the lower bound is inferred from the four calibrated points.

Nonclaim 11.3 (No exact-four theorem). *The statement $|\mathfrak{Q}_5^{\text{SCIC}}| \geq 4$ is strict lower closure. Neither $|\mathfrak{Q}_5^{\text{SCIC}}| = 4$ nor finiteness is claimed.*

11.5 Remaining classification problem

Completing PCR-5A requires executing the family-image classification already fixed by the preceding definitions and theorems:

1. partition every LAW, LOCAL, HID, COMP, SOL, and HYB survivor domain into packet-constant or exact packet-moduli strata;
2. prove the failure boundary of every stratum;
3. calculate all cross-family overlap maps;
4. assemble the resulting stratum atlas.

Only after these steps is the standing-carrier census complete at the classification level defined in section 1.

12 Robustness and falsifiability

12.1 Essential theorem chain

The established implications are

candidate completeness $\rightarrow$ survivor completeness $\rightarrow$ SCIC activation
 $\rightarrow$ generator/action factorization $\rightarrow$ packet completeness
 $\rightarrow$ mixed four-class theorem $\rightarrow$ packet-image representation.

The additional passage from family packet images to a complete stratum atlas is a separate, unresolved classification problem.

12.2 Direct falsifiers

Any of the following would invalidate one of the established claims:

1. a lawful same-target PCR-4 survivor outside the inherited family cover;
2. an admitted standing consequence not factoring through the primitive generator algebra;
3. a failure of the total fail-closed continuation action to recover some $w(u(x))$;
4. two equal SCIC signatures separated by the packet, or two unequal signatures merged by it;
5. a PCR-3 hybrid non-survivor receiving a standing image;
6. a nonzero PCR-4 simultaneous coordinate or cross residual in either explicit mixed witness;
7. failure of either explicit common cone;
8. an error in one of the four deletion-pair calculations.

12.3 The unresolved family stratification is a separate question

An unclassified packet stratum in LAW, LOCAL, HID, COMP, SOL, or HYB would not refute the packet-image representation. It would instead enlarge the still-open stratum atlas. The distinction between exact representation and executed classification therefore prevents an unresolved family sector from being mistaken for a counterexample to the closed quotient theorems.

12.4 Changes that do not affect the theorem

The following do not defeat the established results:

- preferring the lowest-dimensional or fewest-field mechanism;
- identifying carriers merely because their low-energy coefficients agree;
- counting source-family names as quotient classes;
- importing heavy-state, collider, cosmological, nonunitarity, neutrinoless-double-beta, or global $B - L$ data from a richer target;
- treating a finite registry or consistency checker as a semantic proof.

12.5 Structural tests of the mixed constructions

The two mixed points cease to be the same PCR-4 witnesses under any change that alters their one-point simultaneous data:

1. retuning after deletion violates the no-repair diagnostic;

2. promoting the singlet matched output to an independent LAW source violates matching orientation;
3. separate vacua destroy the explicit common cone;
4. a disconnected anomaly spectator changes the source architecture;
5. ignoring null outcomes destroys event completeness;
6. replacing complete-hybrid evaluation by a constituent pass/fail product changes the PCR-4 survivor predicate.

12.6 Negative controls

A source difference creates no class when every packet coordinate agrees. Two inequivalent gluings can collapse when all induced standing values agree. A raw parameter can vary inside a provenance fibre when no standing evaluator changes. Thus the four-class result follows from the specific no-repair responses, not from counting mechanisms.

Theorem 12.1 (Exhaustive failure modes for the established results). *Every faithful challenge to the established theorem chain has one of four dispositions:*

1. *upstream candidate/survivor incompleteness;*
2. *generator/action or packet incompleteness;*
3. *failure of an explicit mixed-survivor construction or deletion value;*
4. *incorrect packet equality or family-cover representation.*

A challenge to the still-open family stratification is not a contradiction of these claims; it belongs to the remaining PCR-5A classification problem.

Proof. The established result is the composition of the upstream cover, packet isomorphism, explicit mixed witnesses, and family packet-image representation. Any counterexample to those claims must break a map in that composition. The stratum atlas is not asserted as a completed theorem here. $\square$

13 Relation to PCR-6

The exact quotient $\mathfrak{Q}_5^{\text{SCIC}}$ already exists by PCR-5 and is represented faithfully here by $\mathfrak{M}_5^{\text{stand}}$. Four classes are additionally separated by explicit standing witnesses. What is not yet available is a complete stratum atlas of the remaining quotient, and that missing classification limits the scope of any subsequent class-by-class physical characterization.

Theorem 13.1 (PCR-6 handoff boundary). *PCR-6 may use the four calibrated classes*

$$q_W, \quad q_{3N}, \quad q_{\text{red}}, \quad q_{\text{syn}}$$

as proved inputs for comparative characterization. A characterization theorem covering the entire carrier domain, however, requires completion of the PCR-5A family stratification and overlap atlas.

Proof. The four classes have explicit survivor witnesses and standing separators. The packet-image representation is exact as a quotient representation but does not enumerate or parameterize every remaining stratum. A global characterization before that classification is complete would risk importing downstream descriptive distinctions into an upstream identity problem that remains unresolved. $\square$

13.1 Four calibrated classes

The first two classes are single-source anchors. The latter two show that source composition introduces standing-distinct redundancy and synergy types. Their class-wide properties can be studied without assuming that they exhaust the quotient.

13.2 Data required for a complete characterization

Once the family atlas is complete, PCR-6 should receive for every stratum or class:

1. the complete standing packet;
2. its exact PCR-4 survivor domain and lawful parameter quotient;
3. its source-provenance fibre;
4. its cross-family overlap maps;
5. minimal standing separators;
6. compatible PCR-5 role-occupancy data;
7. current trace and impossibility status.

Proposition 13.2 (No characterization-to-selection backflow). *A class can be more familiar, simpler, more local, or easier to embed without becoming the realized class. Characterization data have no selector authority unless a new standing trace or impossibility theorem is independently formed.*

Proof. SCIC identity and PCR-5 realization status are fixed before PCR-6. Descriptive predicates do not change either object. $\square$

14 Conclusion

The standing-carrier problem left open by PCR-5 separates naturally into identity representation and family classification. At the identity level, the seven-component packet is complete: the primitive standing-generator algebra factors every admitted evaluator, the continuation action is total and fail closed, and packet equality is equivalent to equality of the full SCIC signature. Consequently,

$$\mathfrak{Q}_5^{\text{SCIC}} \cong \mathfrak{M}_5^{\text{stand}} = \bigcup_{F \in \mathfrak{F}_5^{\text{surv}}} \text{im}(\Xi_5 \circ \iota_F)$$

is an exact representation of the target-relative standing quotient.

The mixed LAW-plus-singlet sector then shows that the two benchmark classes do not exhaust that quotient. The phase-redundant and rank-split synergistic HYB points satisfy the simultaneous PCR-4 survivor conditions at one parameter point and have deletion pairs (1, 1) and (0, 0), respectively. Together with the anchor values (0, 1) and (1, 0), they give

$$|\mathfrak{Q}_5^{\text{SCIC}}| \geq 4.$$

Thus exact two and exact three are excluded within the fixed target and comparison class.

The result does not imply exact four. The general LAW, LOCAL, HID, COMP, SOL, and HYB images can still contain additional packet-constant, discrete, continuous, or categorical strata whenever a standing-visible coordinate varies. Determining those strata and their cross-family overlaps is the remaining classification problem. Until that atlas is complete, $\mathfrak{M}_5^{\text{stand}}$ should be read as an exact packet-image representation rather than as an executed census of every stratum.

The physical conclusion is therefore twofold. One terminal neutrino propagation role admits at least four irreducible same-target standing-carrier realizations, and the full quotient is already represented without ambiguity by complete packet values. At the same time, the realized carrier is not selected by the fixed target, and a complete physical characterization of the entire quotient remains downstream of the unresolved family-stratification problem.

A Notation and type hierarchy

Symbol	Meaning
$A_5 = \mathfrak{S}_v^{(4)}$	Candidate-complete PCR-4/PCR-5 source-carrier survivor family.
Σ_5^{SCIC}	Complete target-authorized standing-continuation signature.
$\mathfrak{Q}_5^{\text{SCIC}}$	Final same-scope standing-carrier quotient.
Ξ_5	Complete seven-component PCR-5A census packet.
$\mathfrak{G}_5^{\text{nf}}$	Raw seven-type census normal form; not itself standing identity.
$\overline{\mathfrak{G}_5^{\text{nf}, \text{stand}}}$	Standing-evaluator quotient of the raw normal-form codomain.
$\text{Act}_5^{\text{nf}}, \overline{\text{Act}_5^{\text{stand}}}$	Total fail-closed raw continuation action and its standing descent.
ι_F	Canonical family-presentation map into A_5 .
$\text{HybDec}_5^{\text{pre}}, \text{HybDec}_5^{\text{surv}}$	Pre-survivor and PCR-4-survivor decorated hybrid groupoids.
$\mathfrak{M}_5^{\text{stand}}$	Exact global packet-image representation object.
$x_{\text{red}}, x_{\text{syn}}$	Redundant and synergistic mixed calibration survivors.

The identity hierarchy is

$$\text{source presentation} \rightarrow \text{inherited carrier} \rightarrow \text{standing carrier} \rightarrow \text{coarse target projection}.$$

Every arrow can forget information; no converse is automatic.

A.1 Standing versus provenance

For $q \in \mathfrak{Q}_5^{\text{SCIC}}$, define

$$\text{Prov}(q) = \{x \in A_5 : [x]_{\text{SCIC}} = q\}.$$

The packet is constant on $\text{Prov}(q)$, while source family, basis, chart, and microscopic parameter data can vary. PCR-6 receives both the class packet and this fibre.

A.2 Deletion conventions

The value one means that the complete ordered role packet remains standing after typed no-repair deletion. The value zero means that at least one required witness or constitutive standing path is destroyed. The fail-closed value $\perp$ denotes an ill-typed diagnostic and is distinct from physical preservation or destruction.

B Detailed mixed-source algebra

B.1 Overall phase invariance

For any $K \in \text{Sym}_3(\mathbb{C})$ and phase $e^{i\phi}$,

$$(e^{i\phi}K)(e^{i\phi}K)^\dagger = KK^\dagger.$$

Thus a common phase changes neither singular values nor the oscillation Hamiltonian built from $KK^\dagger$. This is the algebraic basis of the redundant construction.

B.2 Cube-root identity

With $\omega = e^{2\pi i/3}$,

$$1 + \omega = e^{i\pi/3}, \quad |1| = |\omega| = |1 + \omega| = 1.$$

Therefore K_* , ωK_* , and $(1 + \omega)K_*$ have identical Hermitian squares.

B.3 Rank split

Let $K_* = UDU^T$ with $D = \text{diag}(d_1, d_2, 0)$, $d_1 \neq d_2 > 0$. Writing u_i for the first two columns of U ,

$$K_* = d_1 u_1 u_1^T + d_2 u_2 u_2^T.$$

Each summand has rank one and only one nonzero singular value. Their sum has two independent mode scales. Choosing U non-diagonal in the charged-current basis gives nonalignment.

B.4 Singlet lift

A rank-one term duu^T is realized by one singlet with mass $M > 0$ and Yukawa column $Y = \sqrt{dM} u$. A general symmetric term is realized by the Takagi lift of lemma 8.1.

B.5 Deletion table

Carrier	LAW deletion	new-local deletion	Reason
x_W	0	1	LAW is constitutive; no local source is active.
x_{3N}	1	0	No independent LAW source; singlet path is constitutive.
x_{red}	1	1	Either phase-related mechanism independently realizes the same target.
x_{syn}	0	0	Both rank-one contributions are jointly required for two modes.

B.6 PCR-4 simultaneous HYB residual conditions

The mixed-survivor proofs in section 8 evaluate the complete hybrid point rather than a product of marginal constituent verdicts. To make that evaluation explicit, let

$$R_{\text{sum}} := \kappa_{\text{tot}} - \kappa_W - \kappa_N, \quad R_{\text{match}} := \kappa_N - Y_\nu M_R^{-1} Y_\nu^T.$$

The remaining residual symbols below are the corresponding source-interpreted coordinates of the imported PCR-4 HYB evaluator: $R_{\text{anom}}^{\text{cross}}$ for the cross-anomaly condition, $R_{\text{vac}}^{\text{cross}}$ for common-vacuum compatibility, $R_{\text{bdy}}^{\text{cross}}$ for common boundary/attachment compatibility, and $R_{\text{ext}}^{\text{cross}}$ for any activated-extension relation. They are not new survival conditions; they make explicit the generic HYB coordinates recorded in section I.

For both constructed points, at one common parameter point,

$$R_{\text{sum}} = 0, \quad R_{\text{match}} = 0, \quad R_{\text{anom}}^{\text{cross}} = 0, \quad R_{\text{vac}}^{\text{cross}} = 0, \quad R_{\text{bdy}}^{\text{cross}} = 0, \quad R_{\text{ext}}^{\text{cross}} = 0.$$

The reasons are pointwise rather than existential: the coefficient is defined as the displayed sum; the singlet coefficient is the displayed exact Takagi/one-singlet match; the added fermions are gauge singlets; both source terms are evaluated on the same electroweak vacuum and charged-current boundary; and no optional TOP/EXT/BDY/ENV source is activated. Thus the mixed point does not borrow a pass value from a second parameter point or a disconnected spectator.

PCR-4 HYB coordinate	θ_{red}	θ_{syn}
Target/records	fixed benchmark charged-current boundary and target instrument	same fixed boundary and target instrument
Event-complete response	$\kappa_{\text{tot}} = e^{i\pi/3} K_*$, hence $\kappa_{\text{tot}} \kappa_{\text{tot}}^\dagger = K_* K_*^\dagger$	$\kappa_{\text{tot}} = K_*$ exactly
Support/continuation	one common lepton/Higgs support and one source-to-detector continuation	same
Gauge/anomaly	gauge-invariant LAW term plus gauge-singlet N_R ; $R_{\text{anom}}^{\text{cross}} = 0$	same with one active singlet
Vacuum/deformation	one electroweak branch; $R_{\text{vac}}^{\text{cross}} = 0$	same
Oriented matching	$Y_{\text{red}} M_R^{-1} Y_{\text{red}}^T = \omega K_*$; $R_{\text{match}} = 0$	$Y_{\text{syn}} M^{-1} Y_{\text{syn}}^T = d_2 u_2 u_2^T$; $R_{\text{match}} = 0$
Global source relation	$K_* + \omega K_* = e^{i\pi/3} K_*$; $R_{\text{sum}} = 0$	$d_1 u_1 u_1^T + d_2 u_2 u_2^T = K_*$; $R_{\text{sum}} = 0$
Activated extensions	inactive imported value; $R_{\text{ext}}^{\text{cross}} = 0$	inactive imported value; $R_{\text{ext}}^{\text{cross}} = 0$
Common cone	explicit cone C_{red} below	explicit cone C_{syn} below

B.7 Explicit common-cone witnesses

Let $S_{\text{mix}}^{\text{red}}$ and $S_{\text{mix}}^{\text{syn}}$ denote the complete mixed source actions used as the apex objects. For either point there are source-restriction legs

$$r_W : S_{\text{mix}} \longrightarrow S_W, \quad r_N : S_{\text{mix}} \longrightarrow S_N,$$

an effective leg

$$p_{\text{eff}} : S_{\text{mix}} \longrightarrow \kappa_{\text{tot}}, \quad p_{\text{eff}}(S_{\text{mix}}) = \kappa_W + \kappa_N,$$

and the fixed target-evaluation leg $p_T : S_{\text{mix}} \rightarrow \mathcal{T}_*$. The LAW restriction reads the independent κ_W term; the singlet restriction reads (Y_ν, M_R) and its oriented matching map; and the effective leg is the displayed algebraic sum. Consequently the routes through the source restrictions and the effective sum agree with the direct target leg on the same boundary, vacuum, and parameter point:

$$p_T = \text{Eval}_{\mathcal{T}_*} \circ p_{\text{eff}}, \quad p_{\text{eff}} = \text{Sum} \circ (r_W, r_N).$$

For θ_{red} , $p_{\text{eff}} = e^{i\pi/3} K_*$; for θ_{syn} , $p_{\text{eff}} = K_*$. These equalities exhibit a nonempty cone for each complete hybrid and do not splice marginal witnesses from separate models. The source-to-effective matching leg is oriented only on the singlet branch; the independent LAW branch is never reconstructed from the matched output.

B.8 Why the cube-root construction is exact

The equality of Hermitian squares is pointwise, not perturbative:

$$(e^{i\alpha} K_*)(e^{i\alpha} K_*)^\dagger = K_* K_*^\dagger.$$

Therefore every target record computed from the effective oscillation Hamiltonian is unchanged. The construction does not rely on an approximate small coupling or an unobserved region of parameter space.

B.9 Why the rank split is exact

The target's two phase-frequency intervals can be assigned to d_1^2 and d_2^2 relative to the zero mode. Each rank-one constituent has only one nonzero interval; the sum has both. This makes the two zero deletion responses consequences of the fixed response-mode target, not of a discretionary threshold.

C Family normal-form and image schema

Family	Exact survivor domain	Standing packet burden
LAW	PCR-4 equation (51)	Target channel, LAW deletion, continuation, operator order, matching, failure.
LOCAL	PCR-4 equation (52)	Active representation/rank only when standing-visible; local deletion, matching, boundary, failure.
HID	PCR-4 equation (57)	Dilation/projection, hidden support, continuation, matching, failure after isometry quotient.
COMP	PCR-4 equation (53)	Constituent support, derivation map, residual incidence, matching and failure.
SOL	PCR-4 equation (56)	Persistent profile, topology, support, boundary, deformation, continuation and failure.
HYB	PCR-4 theorem 18.1 survivor pullback	Complete decorated gluing, every higher-arity cross effect, common point/cone, continuation and failure.

For every family the exact image is the image of the complete packet map over the full survivor locus, modulo its lawful parameter groupoid. A finite sample is never a family theorem.

C.1 Packet-stratum report format

Each published family stratum is reported by the tuple

$$(\Theta_{\alpha}^{\text{surv}}, \Gamma_{\alpha}, \Xi_{\alpha}, \text{im } \Xi_{\alpha}, \text{Prov}_{\alpha}, \text{Fail}_{\alpha}).$$

The tuple gives the exact domain, lawful quotient, packet map, standing image, provenance fibre, and failure boundary. It is sufficient for reconstruction without choosing a canonical model.

C.2 Family-image falsifier

A proposed family image is incomplete if one can exhibit a survivor point in the same exact domain whose packet value lies outside the stated image. Sampling or absence of a known example cannot close this burden.

D Dependency structure

The following table summarizes the principal results used in the dependency chain. The compact P5A identifiers are retained only as cross-reference labels for the theorem sequence.

Reference	Result	Mathematical consequence
P5A-01	No-backflow	Target, candidate domain, survivor set, occupancy and SCIC relation remain fixed.
P5A-02N	Primitive generator factorization	Every $w \in \mathcal{W}_5^{\text{stand}}$ factors by structural induction through the seven typed primitive generator families.
P5A-02F	Standing-faithful descent	Raw evaluator-invisible distinctions are removed.
P5A-02A	Continuation reconstruction	Total fail-closed generator action recovers every $w(u(x))$.
P5A-02	Packet completeness	$\Xi_5(x) = \Xi_5(y) \iff x \sim_{\text{SCIC}} y$.

Reference	Result	Mathematical consequence
P5A-04	Family packet-image cover	Six inherited family packet images represent every quotient class.
P5A-11S	HYB survivor restriction	Only PCR-4 surviving decorated hybrids map into A_5 .
P5A-11	Decorated colimit	Complete surviving hybrid packet is decomposition-independent.
P5A-11M	Mixed calibration	Explicit redundant and synergistic HYB points survive; four deletion pairs are realized.
P5A-15R	Packet-image representation	$\mathfrak{M}_5^{\text{stand}} \cong \mathfrak{Q}_5^{\text{SCIC}}$ as an exact representation.
P5A-17X	Four-class lower closure	$ \mathfrak{Q}_5^{\text{SCIC}} \geq 4$; exact two and exact three are false.
P5A-18O	Open family stratification	General six-family packet images are not yet fully stratified.
P5A-19B	PCR-6 characterization boundary	Global characterization requires completion of P5A-18O.

The dependency order is

$$\text{P5A-01} \rightarrow \text{P5A-02N/F/A} \rightarrow \text{P5A-02} \rightarrow \text{P5A-04,11S,11} \rightarrow \text{P5A-11M} \rightarrow \text{P5A-15R,17X},$$

with P5A-18O remaining open before the complete P5A-19 characterization can be formed.

D.1 Dependencies of the four-class theorem

The four-class theorem depends on:

1. strict BASE exclusion after LAW deletion at x_W ;
2. no matching backflow after LAW deletion at x_{3N} ;
3. exact Takagi singlet realization;
4. complete-hybrid evaluation on the explicit θ_{red} and θ_{syn} points, without any constituentwise survivor premise;
5. vanishing intrinsic/cross PCR-4 residuals and explicit nonempty common cones at those points;
6. standing-bearing status of both deletion coordinates.

E Adversarial robustness and falsifiers

The theorem chain admits direct structural challenges. The table records the form a counterexample would have to take and the mathematical consequence if such a counterexample were produced.

Potential challenge	Required counterexample	Consequence
Candidate omission	Same-target survivor outside PCR-3/PCR-4 cover	Reopens upstream completeness.
Packet under-resolution	Standing evaluator not recovered by packet	Refutes packet theorem.
Packet over-resolution	Packet distinction invisible to all standing evaluators	Removed by standing quotient.

Potential challenge	Required counterexample	Consequence
Continuation relabeling	Same outputs after permutation but not pointwise in u	Excluded by the labeled function profile.
HYB survivor leakage	PCR-4 non-survivor receives standing colimit	Excluded by the survivor pullback.
Gluing ambiguity	Same vertices, inequivalent decorated gluings	Retained as distinct gluing objects.
Higher-arity omission	Standing hyperedge not generated pairwise	Retained in complete hypergraph.
Redundant mixed failure	Show phase-redundant hybrid violates a named PCR-4 condition	Removes q_{red} .
Synergistic mixed failure	Show rank-split hybrid violates a named PCR-4 condition	Removes q_{syn} .
Unlawful merge	Identify unequal deletion pairs	Violates witness soundness.
Selector import	Use naturalness, field count, or richer-target evidence	Outside the theorem's scope.
Finite-registry substitution	Treat data files as semantic proof	Does not establish the semantic theorem.

E.1 Failure modes excluded by the theorem chain

The following candidate defects are excluded by the results proved above: omission of the target-facing packet coordinate; substitution of static identity for continuation identity; packet-only hybrid composition; omission of higher-arity gluing; family-presentation domain mismatch; promotion of source coherence to PCR-4 survival; and raw normal-form over-resolution.

F Source register and import boundary

The load-bearing scientific sources are PCR-2 through PCR-5, SCIC, and Anchor-Tensor-Skin. PCR-3 supplies exact source generation; PCR-4 supplies the exact simultaneous-survivor loci; PCR-5 supplies compatible occupancy, SCIC activation, and standing deletion semantics. Standard EFT and singlet constructions are used only as explicit witnesses [1, 8–11].

Source	Imported object
PCR-2	Complete target-facing interface, target-use/evaluator exhaustion, continuation and matching typing.
PCR-3	Source ontology, finite coherent hypergraphs, family templates, candidate quotient and parametric closure.
PCR-4	Exact family evaluators, survivor loci, common-cone relations, hybrid survivor locus and exact quotient.
PCR-5	Universal compatible occupancy, SCIC scope, deletion response and two benchmark anchors.
SCIC	Unique target-relative standing identity quotient and no hidden classifier.
ATS	Carrier descent and elimination of standing-inert descriptive refinement.

Any theorem requiring a new empirical record, background, operator category, global form, or target coordinate belongs to a successor target and cannot alter the present quotient.

F.1 Standard witness sources

Weinberg’s dimension-five operator supplies the independent LAW witness, while the singlet construction supplies the local/hidden source and Takagi lift [8, 10, 11]. Stinespring/Kraus results support the distinction between reduced channel identity and retained dilation identity [1, 9]. These sources do not supply the AASC completeness or SCIC theorems.

G Scope boundaries and nonclaims

Nonclaim G.1 (No multiple terminal roles). *PCR-5A counts carrier identities of one terminal complete-carrier role.*

Nonclaim G.2 (No exact finite cardinality). *The theorem proves $|\mathfrak{Q}_5^{\text{SCIC}}| \geq 4$, not $|\mathfrak{Q}_5^{\text{SCIC}}| = 4$ and not finiteness.*

Nonclaim G.3 (No executed exact family census). *The union $\mathfrak{M}_5^{\text{stand}} = \bigcup_F \text{im}(\Xi_5 \circ \iota_F)$ is an exact packet-image representation of the quotient. The general family images are not claimed to have been completely decomposed into exact packet strata.*

Nonclaim G.4 (No empirical selection). *No class is selected as realized. The result contains no probability, likelihood, naturalness ranking, or preferred mechanism.*

Nonclaim G.5 (No low-energy identity). *Equal low-energy coefficients do not identify source carriers or standing packets.*

Nonclaim G.6 (No raw source counting). *Family names, node counts, and parameter values are not class labels unless they alter admitted standing evaluations.*

Nonclaim G.7 (No unrestricted ultraviolet census). *The classification is relative to the PCR-3 comparison class and fixed PCR target.*

Nonclaim G.8 (No Dirac particle–antiparticle theorem). *The presence of several mathematical branches does not imply charge conjugation, energy-sign pairing, or simultaneous realization.*

Nonclaim G.9 (No complete PCR-6 handoff). *The four calibrated classes may be characterized individually, but a characterization of the entire quotient requires completion of the PCR-5A family-stratification problem.*

Nonclaim G.10 (No assertion that all four classes are realized). *The theorem proves four lawful standing classes, not simultaneous population of four carrier architectures in Nature.*

Nonclaim G.11 (No claim that a mixed source is ontologically fundamental). *A mixed class is a standing realization type at the fixed target. PCR-5A does not decide whether one source term is more fundamental than another at a richer target.*

H Reproducibility and certification boundary

The supplementary archive contains machine-readable theorem, family-image, mixed-calibration, endpoint, source, and adversarial-test records. Their purpose is to support provenance, internal consistency, and reproducibility; they are not substitutes for the semantic arguments in the manuscript.

The accompanying consistency checks verify:

1. theorem identifiers and dependency edges;
2. the four deletion-pair entries and their pairwise distinction;
3. family coverage and overlap symmetry;

4. survivor-only HYB category typing;
5. consistency of the lower bound $|\mathfrak{Q}_5^{\text{SCIC}}| \geq 4$;
6. absence of unresolved references, citations, or box warnings;
7. deterministic archive hashes.

H.1 Semantic proof and finite certification

The CSV and JSON records encode theorem identifiers, the four deletion pairs, family coverage, and endpoint consistency. A consistency checker can detect stale entries, a missing family row, or contradictory endpoint metadata. It cannot prove source completeness, PCR-4 survival, packet completeness, or the mixed common-cone constructions. Those claims are established only by the mathematical arguments given above.

I Exact PCR-4 survivor domains used by the census

The global quotient is exact only because its domain is the previously proved PCR-4 survivor object, not a finite list of named models. This appendix records the family domains used in the packet maps. The notation follows the PCR-4 publication manuscript [4].

I.1 Universal family evaluator

For every PCR-3 family F , PCR-4 constructs a source-interpreted evaluation map

$$\widehat{\Phi}_F : \Theta_F^{\text{part}} \longrightarrow \mathcal{A}_4^{\text{sem}}$$

and a closed semantic survival locus $\mathcal{Z}_4^{\text{sem}} \subseteq \mathcal{A}_4^{\text{sem}}$. The exact family survivor domain is

$$\Theta_F^{\text{surv}} = \widehat{\Phi}_F^{-1}(\mathcal{Z}_4^{\text{sem}}). \quad (5)$$

The condition is simultaneous: intrinsic target equations, all family-specific nonemptiness requirements, every cross-signature residual, and common-cone existence are evaluated at one source point.

Theorem I.1 (Imported survivor-domain exactness). *Equation (5) is equivalent to positivity under every PCR-4 semantic survival evaluator. Its complement carries a nonempty typed failure set containing every failed intrinsic coordinate, every nonzero relation residual, common-cone status, and any target/source-update disposition.*

Proof. This is PCR-4 Theorem 15.1 and Corollary 15.2, specialized to the fixed PCR-5A target and comparison class. PCR-5A changes neither the evaluator nor the survivor relation. $\square$

I.2 Common simultaneous conditions

Every principal family locus includes the following conditions, interpreted through that family's source data:

1. complete target-record agreement and an event-complete propagation channel;
2. charged-current nonalignment and two nuisance-separated visible response-mode classes;
3. nonempty causal/support continuation on the fixed boundary and target domain;
4. one admissible vacuum/deformation branch;
5. every required oriented matching or retained-resolution map;
6. vanishing family relation residual $\mathcal{R}_F(\theta)$;
7. a nonempty common-cone set $\text{Cone}_F(\theta)$.

The family-specific data below supplement, rather than replace, these simultaneous conditions.

I.3 LAW and LOCAL

For LAW, the source point contains a source-coherent process/law object on the existing carrier substrate. The exact locus is PCR-4 equation (51): in addition to the common conditions, the process is gauge-covariant on the charged-current boundary, its channel is normalized and completely positive, and the LAW vacuum and matching sets are nonempty.

For LOCAL, let $\mathcal{A}_{\text{loc}}(\theta)$ be the set of gauge-invariant source attachments from the fixed lepton/Higgs support to active local nodes. PCR-4 equation (52) is

$$\Theta_{\text{LOCAL}}^{\text{surv}} = \{ \theta : \mathcal{A}_{\text{loc}}(\theta) \neq \emptyset, \mathcal{A}_{\text{LOCAL}}^{\text{anom}}(\theta) = 0, \text{ and all common simultaneous conditions hold} \}.$$

Representation, charge, anomaly, channel, vacuum, and matching data are evaluated on the same active local content.

I.4 COMP, SOL, and HID

For COMP, let $C_{\text{comp}}(\theta)$ contain a source-coherent derivation/binding map, a target-preserving matching map, and one shared constituent/derived anomaly-vacuum witness. PCR-4 equation (53) gives

$$\Theta_{\text{COMP}}^{\text{surv}} = \{ \theta : C_{\text{comp}}(\theta) \neq \emptyset, \mathcal{A}_{\text{COMP}}^{\text{anom}}(\theta) = 0, \text{ and all common simultaneous conditions hold} \}.$$

For SOL, the family datum $\mathcal{S}_{\text{sol}}(\theta)$ includes a persistent localized profile, its internal response, boundary attachment, common vacuum, deformation/endability, and matching data. PCR-4 equation (56) is the corresponding common-condition locus with $\mathcal{S}_{\text{sol}}(\theta) \neq \emptyset$ and vanishing anomaly residual.

For HID, $\mathcal{H}_{\text{hid}}(\theta)$ is the set of source-coherent dilations or hidden process objects whose support, charges, anomaly data, vacuum branch, and matching map lie in one active trace. PCR-4 equation (57) is the corresponding common-condition locus with $\mathcal{H}_{\text{hid}}(\theta) \neq \emptyset$ and vanishing hidden-sector anomaly residual.

I.5 HYB and pure-sector reductions

The exact hybrid locus is

$$\Theta_{\text{HYB}}^{\text{surv}} = \widehat{\Phi}_{\text{HYB}}^{-1}(\mathcal{Z}_4^{\text{sem}}),$$

where $\widehat{\Phi}_{\text{HYB}}$ evaluates every cross anomaly, cross vacuum relation, mixed matching map, activated-extension coordinate, higher-arity incidence, and common-cone condition of the complete glued architecture. Constituent verdicts are not multiplied.

PCR-4 separately proves that the strict BASE stratum and the pure TOP, EXT, BDY, and ENV strata are empty. Any surviving contribution of those types occurs in an already counted principal-family overlap or in $\Theta_{\text{HYB}}^{\text{surv}}$. Hence

$$A_5 \cong (\Theta_{\text{LAW}}^{\text{surv}} \sqcup \Theta_{\text{LOCAL}}^{\text{surv}} \sqcup \Theta_{\text{COMP}}^{\text{surv}} \sqcup \Theta_{\text{SOL}}^{\text{surv}} \sqcup \Theta_{\text{HID}}^{\text{surv}} \sqcup \Theta_{\text{HYB}}^{\text{surv}}) / \simeq_{\text{inst}}.$$

This is the candidate-domain upper-closure theorem used by theorem 5.2.

J Packet maps and unresolved family stratification

For each principal family,

$$\Xi_F = \Xi_5 \circ \iota_F : \Theta_F^{\text{surv}} / \Gamma_F \longrightarrow \mathcal{Y}_T \times \mathcal{Y}_R \times \mathcal{Y}_D \times \mathcal{Y}_C \times \mathcal{Y}_M \times \mathcal{Y}_B \times \mathcal{Y}_F.$$

The codomains may be symbolic, noncompact, or categorical.

Family	Source-normal-form input	Packet coordinates requiring classification
LAW	Process/law object on existing substrate; source attachment and operator ancestry.	target/evaluator profile; all LAW-kind deletions; continuation; matching/order; boundary; failure.
LOCAL	Active primitive nodes, representations, multiplicity, attachment rank, mass/Yukawa data.	role incidence; kindwise deletion; continuation/threshold; matching; boundary; anomaly/vacuum/failure.
HID	Dilation/enlarged-state-space object, projection, hidden support, isometry class.	visible profile; hidden deletion; projected continuation; matching; hidden ownership; failure/undefinedness.
COMP	Constituent diagram, derivation/binding map, derived node, residual incidence.	constituent deletion/retention; role incidence; continuation; matching; boundary; failure.
SOL	Persistent localized profile, topology/support, deformation/endability.	target response; profile deletion; persistence continuation; matching; bulk/boundary attachment; failure.
HYB	Complete surviving decorated hypergraph with every typed hyperedge and common cone.	all seven coordinates, including cross-effect-induced and higher-arity continuation/failure values.

Theorem J.1 (Family packet-map fibre exactness). *For every principal family F , the fibres of Ξ_F are exactly the restrictions of SCIC classes to that family presentation groupoid. Cross-family equal packet values generate neither a finer nor a coarser relation than $\sim_{\text{SCIC}}$.*

Proof. Family parameter descent gives representative independence and [theorem 4.4](#) gives both implications between packet equality and SCIC equality. $\square$

J.1 Data required for an executed stratum

An executed family stratum must be reported by

$$\mathfrak{S}_\alpha = (\Theta_\alpha^{\text{surv}}, \Gamma_\alpha, \Xi_\alpha, \text{im } \Xi_\alpha, \text{Prov}_\alpha, \partial \text{Fail}_\alpha).$$

This tuple is fully determined for the four calibrated singleton strata, but the remaining family domains have not yet been partitioned into such tuples.

J.2 Four calibrated zero-dimensional strata

$$\mathfrak{S}_W = \{s_W\}, \quad \mathfrak{S}_{3N} = \{s_{3N}\}, \quad \mathfrak{S}_{\text{red}} = \{s_{\text{red}}\}, \quad \mathfrak{S}_{\text{syn}} = \{s_{\text{syn}}\}.$$

Their deletion-coordinate projections are the four points of $\{0, 1\}^2$. These singleton strata need not exhaust any family image.

J.3 Remaining family-stratification problem

Family	Classification result still required for an executed exact census
LAW	exact partition of all standing-visible continuation, operator-order, matching, boundary and failure variants;
LOCAL	exact partition by standing-visible representation/rank/multiplicity/threshold data;

Family	Classification result still required for an executed exact census
HID	exact minimal-dilation/support/projection packet strata and LOCAL/HID overlap atlas;
COMP	exact constituent/derivation packet strata and overlap dispositions;
SOL	exact persistence/topology/endability packet strata;
HYB	exact classification of all remaining decorated-hybrid packets, including higher-arity cross effects beyond the two calibrated points.

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